Master Thesis

Zero-Shot Segmentation and Classification of Urban Land Types Around Greenery Using Foundation Models (SAM and CLIP) with Spatial Buffer Analysis

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Erlangen, 30$^{\text{th}}$ April 2026.

Puneetha Dharmapura Shrirama

Abstract

Urban trees provide essential ecosystem services, and characterising the land composition surrounding individual trees is relevant for urban planning and environmental monitoring. Existing supervised approaches for aerial land-cover classification depend on large annotated datasets and fail to generalise across cities, sensors, and seasons, while no prior work has connected zero-shot foundation model outputs to GIS-based spatial analysis for tree-centred land composition estimation. This thesis therefore investigates whether foundation models can estimate multi-class land composition around individual urban trees without labelled data, examining the effect of multi-scale segmentation fusion, aerial-specific prompt design, buffer radius, and cross-dataset transfer on pipeline performance. A pipeline combining the Segment Anything Model for segmentation, CLIP for classification, and GIS buffer analysis was developed and evaluated on 95 tree locations in Erlangen, Germany, across three concentric buffer rings (0–2.5 m, 2.5–5.0 m, 5.0–7.5 m). Multi-scale segmentation fusion improved mean pixel coverage from 70.5% to 96.8%, and aerial-specific prompt refinement improved Vegetation F1-scores to 0.68–0.73 and Road F1-scores to 0.26–0.51 across rings, while Building classification remained consistently weak. Applied without modification to Aerial 2025 and Google Satellite imagery, the pipeline transferred successfully to stable vegetation-dominated scenes, with Google Satellite achieving Vegetation F1 up to 0.82, while construction sites and strong shadows remained systematic failure modes. The results demonstrate that zero-shot foundation model pipelines can support scalable urban tree monitoring without manual annotation, offering a practical tool for cities requiring land composition estimates across large tree inventories. The code for this thesis is publicly available at `https://github.com/puni-ram48/urban-tree-landcover-zeroshot`.

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The computational experiments in this thesis were carried out using HPC resources provided by the Erlangen National High Performance Computing Center (NHR@FAU) of the Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU). NHR funding is provided by federal and Bavarian state authorities. NHR@FAU hardware is partially funded by the German Research Foundation (DFG) – 440719683.

Finally, I want to thank my family and friends for being there throughout this journey. Their encouragement and belief in me meant more than I can put into words, especially during the moments when things felt uncertain.

Contents

1 INTRODUCTION

1.1 Motivation and Problem Statement

Urban tree health is influenced by environmental factors such as soil type, temperature, and water availability. Healthy urban trees also provide important ecosystem services, including cooling the environment, improving air quality, and supporting biodiversity. Because these benefits depend on the health of individual trees, it is important to understand the conditions that affect their ability to access water. One key factor is the type of land around each tree. Trees surrounded by vegetation receive more water and they are able to withstand the drought conditions better. However, trees located near pavements or roads receive less water and they experience higher stress levels [1]. Therefore, understanding the land around trees is essential for explaining tree health and guiding future urban planning.

High-resolution aerial images are used to measure the land-cover near each tree, including roads, buildings, vegetation, and water bodies [2]. In earlier approaches, this information was collected through traditional field surveys or manual Geographic Information System (GIS) based assessments. As a result, the process was highly labour-intensive and time-consuming. Manual annotation of aerial images also requires a great deal of effort, and factors such as shadows, lighting, or seasonal changes can reduce the consistency of measurements [3]. Even when annotated datasets exist, these variations make it difficult to reproduce results across different cities. At the same time, rapid urbanization and climate change are increasing the need for large scale assessments of urban tree health to support sustainable city planning.

To better understand these challenges, a real-world example highlights why analysing the land surrounding urban trees is important. This is especially visible in Karlsruhe, where the city struggles to keep track of its extensive urban tree population. Karlsruhe has approximately 354,000 urban trees, yet only about 136,000 are actively monitored in the city's tree register. Planting a single tree costs around €2,000. During 2018, nearly 30% of young trees planted between 2014 and 2017 died because of drought and heatwave [4]. Managing tree health at this scale requires current information of each tree, which is impossible to collect manually across hundreds of thousands of sites. These realities highlight why automated analysis of aerial images is necessary. By automatically segmenting and classifying land-cover features in aerial imagery, these methods can provide the spatial information needed to assess urban tree health at scale without requiring manual annotation. Similar challenges exist across German cities, including Erlangen, where this study is conducted. This study uses tree locations sourced from the Agvolution Climavi City platform, an IoT-based urban tree monitoring system that deploys soil moisture and temperature sensors at individual tree locations across European cities. The sensor deployment coordinates provide a real-world spatial reference for land use assessment, grounding the analysis in an operational monitoring context rather than arbitrarily selected locations.

1.2 Related Work

1.2.1 Traditional and Classical Machine Learning Approaches

Early urban land cover classification was dominated by object-based image analysis(OBIA), where the image was first partitioned into meaningful objects. These objects were then described using handcrafted features such as colour, texture, and shape before being used by rule-based decision systems. Researchers reported that when these rules were carefully tuned to a specific training area, classification accuracy could reach as high as 96% [3]. This indicated that the method can be effective, but only when applied in highly controlled settings. Furthermore, Zou et.al. showed that incorporating LiDAR-derived height information significantly improves the separation of structurally similar classes such as buildings, trees, and low vegetation. However, these improvements require extensive manual effort. The segmentation parameters, feature thresholds, and rule hierarchies had to be iteratively adjusted and validated. Even after finetuning, the rule sets often lost 5–10% accuracy when applied to images from a different sensor [5]. Furthermore, misclassification occurred frequently in shadow regions, class boundaries, and in spectrally ambiguous areas. As a result, these observations indicate that rule-based OBIA lacked robustness and generalization capacity, performing reliably only in low-variance and narrowly defined data domains.

To address the drawback of handcrafted rules, classical machine learning classifiers were gradually introduced into the OBIA pipeline. Comparative studies show that these models can achieve strong performance when provided with high quality segmented objects and well engineered features. Qian et al. found that probabilistic and margin-based classifiers outperform simpler tree-based and distance-based methods, achieving accuracies above 90% with sufficient training data and appropriate parameter tuning [6]. Jozdani et al. further demonstrated that although neural network models can outperform classical algorithms, the margin of improvement is often small and not statistically significant, indicating that segmentation quality and feature design still dominate performance [7]. Crucially, these models remain sensitive to training sample size, parameter settings, and the characteristics of the segmentation process, inheriting many of OBIA weakness. As a result, classical machine learning approaches offer more flexibility than rule-based systems but still struggle to generalize across sensors, seasons, and cities. This motivates the transition toward end-to-end deep learning methods, which is capable of learning features directly from images.

1.2.2 Deep Learning for Aerial Image Segmentation

The shift to deep convolutional neural networks fundamentally changed aerial image segmentation. Rather than relying on hand-crafted features, fully convolutional architectures enabled end-to-end pixel-wise labelling [8, 9], eliminating the feature engineering bottleneck. However, deeper networks progressively lost spatial detail, undermining the precise boundary delineation that segmentation requires. Encoder-decoder architectures addressed this by recovering spatial information through skip connections in explicit decoder pathways. U-Net-style variants of this design proved especially effective for structured features like buildings and roads [10, 11, 12]. Despite these advances, most approaches still treated each semantic class independently. Multi-task learning challenged this by incorporating geographic regularization to exploit structural relationships between classes [13, 14], showing that jointly modelling class dependencies could yield further accuracy gains.

Transformer-based models further enhanced the ability to capture long-range spatial relationships, with SegFormer variants and hybrid attention architectures consistently outperforming CNN-based methods in aerial and Unmanned Aerial Vehicle (UAV) segmentation tasks [15, 16, 17, 18]. Studies have shown that these improvements hold true across different sensor types and image resolutions [19, 20]. Additionally, researchers have developed methods to reduce the labelling burden, such as using pseudo-labels from OpenStreetMap or bounding-box-based supervision, which reduces the need for full pixel-level annotations [21, 22, 23]. Furthermore, some studies have demonstrated that segmentation outputs can be directly converted into area measurements, making them valuable for urban planning [24].

Despite all these advancements, one major issue remains unresolved namely domain shift. Models trained on one dataset often struggle when applied to data from a different city, sensor, or viewpoint. For example, Generative Adversarial Network (GAN) based domain adaptation was able to improve segmentation accuracy from 35% to 52% when transferring between aerial datasets, but it still required labelled source data to function [25]. Similarly, achieving accurate UAV vegetation classification required feature engineering specifically tailored to that dataset [26]. This is a fundamental problem, not a minor one. Supervised models are trained to learn from a specific data distribution, and aerial images can vary greatly across different locations, seasons, and sensors. A pilot study conducted prior to this thesis highlighted this issue: SegFormer [27], trained on street-level Cityscapes data [28] , struggled with road and vegetation classification in aerial images because the top-down perspective was outside of its training scope [29]. This clearly demonstrates that methods that do not rely on labelled data are essential for scaling urban land classification reliably, and this is where foundation models become crucial.

1.2.3 Foundation Models for Zero-Shot Segmentation and Classification

Foundation models move away from the traditional idea of training a separate network for each task. Instead, models such as the Segment Anything Model (SAM) [30], Contrastive

Language–Image Pretraining (CLIP) [31], and self-distillation approaches like DINO [32] are trained on massive and diverse datasets to learn general visual and semantic representations that can be reused across tasks without task-specific supervision. For remote sensing, this naturally raises the question of how well such general-purpose models transfer to aerial imagery, and under which conditions their zero-shot capabilities start to break down.

In the vision language direction, CLIP and its remote-sensing variants have been successfully adapted for zero-shot scene classification. RS-CLIP, RemoteCLIP, GeoRSCLIP, and SkyCLIP report strong zero-shot performance on benchmarks such as AID, UC Merced, and NWPU-RESISC45, where each aerial tile is assigned a single label like dense residential, forest, or highway [33, 34, 35, 36]. In these setups, the model encodes the whole image once and compares it to a set of text prompts and the class with highest similarity becomes the image-level label. This demonstrates that CLIP-style models can capture high-level semantics in remote sensing scenes, but it does not address the more fine-grained problem of distinguishing multiple land-cover categories within the same image. Moreover, several studies show that CLIP's reliability degrades when prompts are poorly formulated or when categories are fine-grained and visually similar [37, 38], which is particularly problematic for separating classes such as roads, bare soil, and certain building materials in aerial images.

In the segmentation direction, SAM has been evaluated and adapted for remote sensing tasks such as road and building extraction [39, 40]. Lightweight adapters, point-based fine-tuning, and text prompts can improve its performance on specific datasets, but these adaptations rely on supervised training and therefore sacrifice the strict zero-shot property. Without any fine-tuning, prompt-engineered SAM can still be used in a training-free way.For example, SAM has been used to refine building masks from pretrained CNNs [41] and to delineate water bodies in high-resolution imagery [42]. However, its performance remains inconsistent across land-cover types and sensor resolutions, with segmentation quality strongly affected by the scale and structural complexity of urban features [14]. In practice, SAM may over-segment homogeneous regions while merging visually mixed areas into a single mask. These limitations motivate the use of open-vocabulary segmentation [43], where vision-language models assign semantic labels to region proposals or segmentation masks rather than entire images. However, most approaches in this category are evaluated on natural image datasets and standard segmentation benchmarks, rather than high-resolution aerial imagery with spatially explicit analysis requirements.

A natural next step has been to combine SAM and CLIP into a joint zero-shot pipeline, letting CLIP handle semantics and SAM provide spatial support. SAM-CLIP is one approach that explores this idea by aligning visual and language representations, enabling prompt-based segmentation without separate inference pipelines [44]. Related pipelines have been explored in remote sensing for unsupervised building extraction [45], zero-shot anomaly segmentation [46], and text-driven object segmentation using Grounding DINO as a language-guided detector [32]. These works demonstrate that label-free analysis of aerial scenes is feasible across several tasks.However, existing approaches typically focus on image-level classification or single-class segmentation tasks, and do not explicitly target dense multi-class land-cover assignment at the segment level within a single aerial image under a fully zero-shot remote sensing setting. Furthermore, they do not integrate these predictions into a GIS-based spatial buffer analysis framework for urban ecological interpretation. This is the focus of this thesis.

1.2.4 GIS-Based Spatial Buffer Analysis

Producing accurate land cover maps is only part of the problem. To turn those maps into practical information about individual urban trees, a spatial analysis step is needed that can quantify the land composition within a defined distance around each tree. Buffer-based land composition analysis is a well-established method for this kind of question. Studies have shown that measuring the proportion of different land-cover types within circular zones around points of interest reliably captures how the surrounding environment shapes local conditions [2, 47]. The computational side is also well covered, with work on large-scale buffer generation and distributed zonal statistics demonstrating that pixel counting within buffer polygons can be performed efficiently even across millions of spatial features [48, 49].

The key gap is that all existing applications of buffer-based land composition analysis start from a supervised land cover map [2, 48, 47, 49]. The labelling dependency that constrains supervised segmentation methods therefore carries over directly into the spatial analysis. The usefulness of the buffer statistics depends entirely on the quality of the upstream classification, which in turn depends on labelled training data. No prior work has connected zero-shot foundation model outputs to GIS buffer analysis to compute multi-class land composition around urban greenery without labelled data, and it is this gap that the present thesis addresses.

1.3 Research Gap

The literature review identifies three fundamental gaps.

First, supervised methods achieve high accuracy on benchmark datasets but fail to generalize across cities due to domain shift. Transfer learning and pseudo-labelling reduce but do not eliminate dependence on labelled data.

Second, existing foundation model applications in remote sensing operate at image level, assigning a single semantic label per scene. No prior work applies CLIP at segment level to classify multiple co-occurring land use classes within a single aerial image, nor integrates multi-class foundation model outputs with GIS-based buffer analysis to quantify land composition around spatial point features.

Third, GIS buffer analysis methods are well-established for quantifying surrounding land composition, but all existing applications begin with supervised land cover maps, thereby inheriting the labelling dependency that limits supervised approaches.

These gaps motivate the need for a framework that connects zero-shot foundation models with GIS-based spatial analysis for urban land composition assessment..

1.4 Thesis Purpose and Contributions

This thesis develops a zero-shot urban land use classification pipeline that integrates SAM-based segmentation, CLIP-based classification, and GIS-based spatial buffer analysis to quantify land use composition around urban trees using high-resolution aerial imagery from Erlangen.

Four research questions guide the experimental design:

RQ1: How does multi-scale segmentation fusion influence spatial coverage and structural consistency in aerial imagery compared to single-scale segmentation?

RQ2: How does aerial-specific prompt design influence land-use classification performance in a zero-shot setting?

RQ3: How does the choice of buffer radius affect the stability of land-use composition estimates around urban trees, and how does class-wise reliability vary across scales?

RQ4: How robust is the proposed zero-shot pipeline when applied to imagery from different years, seasons, and sensors without parameter modification?

Key contributions of this thesis are:

- A zero-shot urban land composition framework that integrates SAM-based segmentation, CLIP-based classification, and GIS buffer analysis for tree-centred spatial evaluation without labelled training data.
- A multi-scale segmentation fusion strategy that improves spatial coverage and reduces fragmentation in complex urban aerial scenes.
- A prompt design strategy tailored for aerial imagery that improves CLIP-based classification performance across heterogeneous land-cover classes.
- A buffer-based spatial evaluation methodology that enables quantitative comparison of land composition across multiple spatial scales and datasets.

1.5 Thesis Structure

This thesis comprises six chapters. Chapter 1 introduces the research context, motivation, and research questions. Chapter 2 presents the theoretical background on SAM, CLIP, and GIS-based spatial analysis. Chapter 3 describes the methodology, including dataset characteristics, pipeline design, and evaluation strategy. Chapter 4 presents the experimental results across segmentation, classification, and buffer evaluation stages. Chapter 5 interprets the findings in relation to the four research questions and addresses limitations of the proposed methodology. Chapter 6 summarizes the main conclusions, outlines directions for future work, and reflects on the broader implications of zero-shot land use classification for urban tree monitoring.

2 THEORETICAL BACKGROUND

2.1 Foundation Models

Foundation models are large neural networks pre-trained on broad and diverse datasets using self-supervised or weakly supervised objectives, enabling them to be reused across many different tasks without being designed for any single application. The term was introduced by the Stanford Center for Research on Foundation Models [50].

Several characteristics distinguish foundation models from earlier generations of machine learning systems [50]. First, they are trained on large corpora spanning multiple modalities such as text, images, audio, or video, enabling the learning of general-purpose representations that capture high-level structure in the data. Second, these models typically contain hundreds of millions to billions of parameters, reducing the dependency on manually annotated labels. Third, once pre-trained, foundation models can be used in zero-shot or few-shot settings, where useful performance is obtained on tasks never explicitly seen during training by changing the input prompt or providing a small number of examples.

In contrast, traditional supervised learning follows a task-specific workflow in which a model is trained from scratch on a labelled dataset prepared for a single task [24, 19, 15, 16, 10, 11]. The performance of such models is determined by the size and quality of the task-specific training set, and deploying a new model for each task requires the collection of new labels and repetition of the full training pipeline [24, 10]. With foundation models, computational and data resources are concentrated into a single large pre-training phase, after which the resulting model is reused across many downstream tasks with relatively little additional effort [50].

Foundation models have been applied across a broad range of computer vision tasks, including image segmentation, object detection, visual question answering, and image-text retrieval, demonstrating strong generalization across domains without task-specific retraining.

2.2 Segment Anything Model

2.2.1 Architecture Overview

The Segment Anything Model (SAM) is a prompt-based segmentation model capable of generating object masks for a wide range of prompts and image types [30]. Its architecture comprises three main components: an image encoder, a prompt encoder, and a mask decoder, as illustrated in Figure 2.1. The image encoder processes the input image into a dense feature representation; the prompt encoder converts user-specified prompts into embeddings compatible with those features; and the mask decoder combines both to predict one or more segmentation masks.

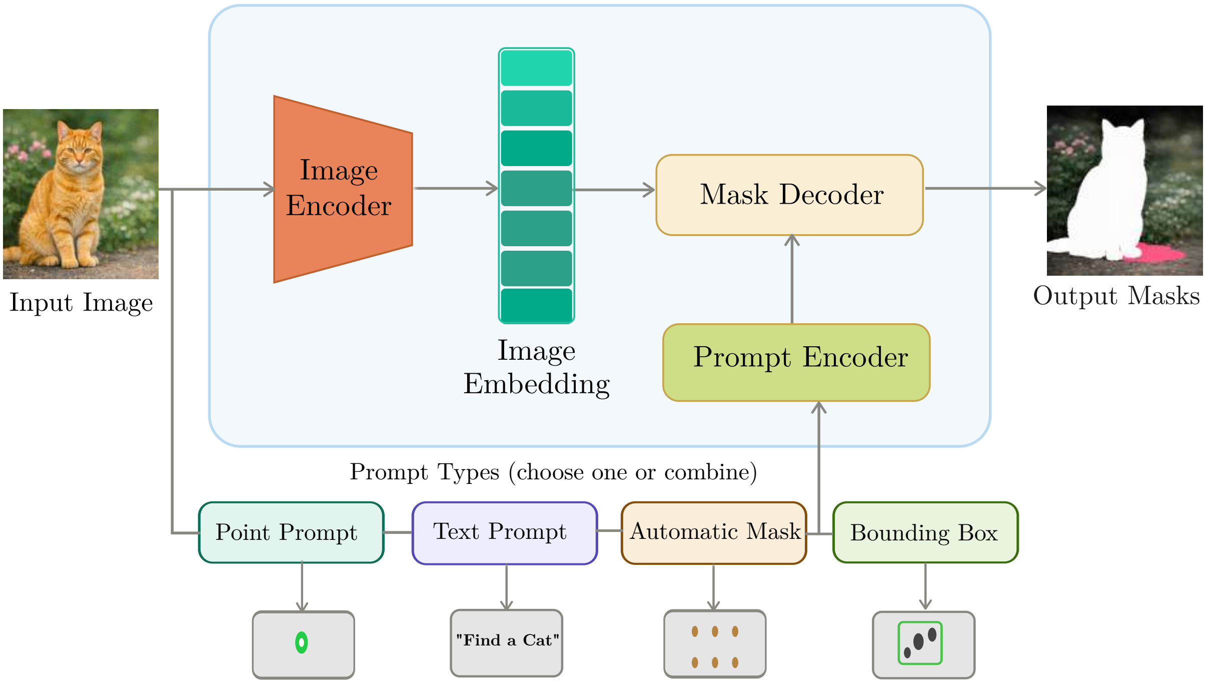

Figure 2.1: Schematic overview of the SAM architecture comprising an image encoder, prompt encoder, and mask decoder, illustrating the four supported prompt types and their corresponding outputs [30]

SAM was trained on the SA-1B dataset, which contains approximately 11 million images and more than one billion segmentation masks, assembled using a data engine that combines model-assisted annotation and automatic mask generation [30]. This large and diverse training corpus enables strong zero-shot generalization across a wide range of prompts and image distributions, allowing the model to be applied to new datasets without task-specific retraining.

2.2.2 Image Encoder

The image encoder in SAM is based on a Vision Transformer (ViT) that maps the input image to a dense feature representation [30]. The image is first divided into a regular grid of non-overlapping patches, each of which is projected into a vector representing its visual content. These patch embeddings are then processed by a transformer encoder that applies self-attention across all patches, allowing information to be shared globally and producing features that capture both local detail and global structure. The output of the image encoder is a compact spatial embedding of the entire image. This embedding is computed once per image and reused for all subsequent prompt evaluations, decoupling the computationally expensive encoding step from the lightweight decoding step and enabling efficient processing of multiple prompts on the same image.

2.2.3 Prompt Encoder

The prompt encoder converts user-specified inputs into embeddings that can be combined with the image features during mask prediction [30]. SAM supports two categories of prompts: sparse and dense. Sparse prompts including points, bounding boxes, and text are represented by their coordinates or token sequences and encoded using positional embeddings combined with learned type embeddings for each prompt category. Dense prompts, such as rough mask inputs, are embedded using convolutional layers and spatially aligned with the image feature map before being passed to the decoder.

2.2.4 Mask Decoder

The mask decoder takes the image embedding from the encoder and the prompt embeddings from the prompt encoder and produces segmentation masks [30]. It is implemented as a lightweight transformer-based module that attends to both the image features and the prompt tokens to predict binary masks over the image. For each set of prompt embeddings, the decoder predicts one or more binary masks together with a confidence score estimating prediction quality. When a prompt is ambiguous, multiple candidate masks are returned to allow downstream selection or post-processing. In automatic mask generation mode, the decoder is applied repeatedly to a dense grid of point prompts sampled across the image, and the resulting masks are filtered by quality thresholds and non-maximum suppression to produce a final set of non-overlapping object segments.

2.3 Contrastive Language-Image Pre-Training

2.3.1 Architecture Overview

CLIP is a vision-language model that learns a joint embedding space for images and natural language descriptions [51]. It comprises two components: an image encoder and a text encoder, as illustrated in Figure 2.2. The image encoder maps an input image to a normalised feature vector, and the text encoder maps a natural language description to a normalised feature vector of the same dimension, both projected into a shared embedding space.CLIP was pre-trained on a dataset of 400 million image-text pairs collected from the internet, referred to as WebImageText (WIT) [31]. This large and diverse training corpus spans a broad range of visual concepts and natural language descriptions, enabling the model to learn general visual-semantic representations without manual annotation.

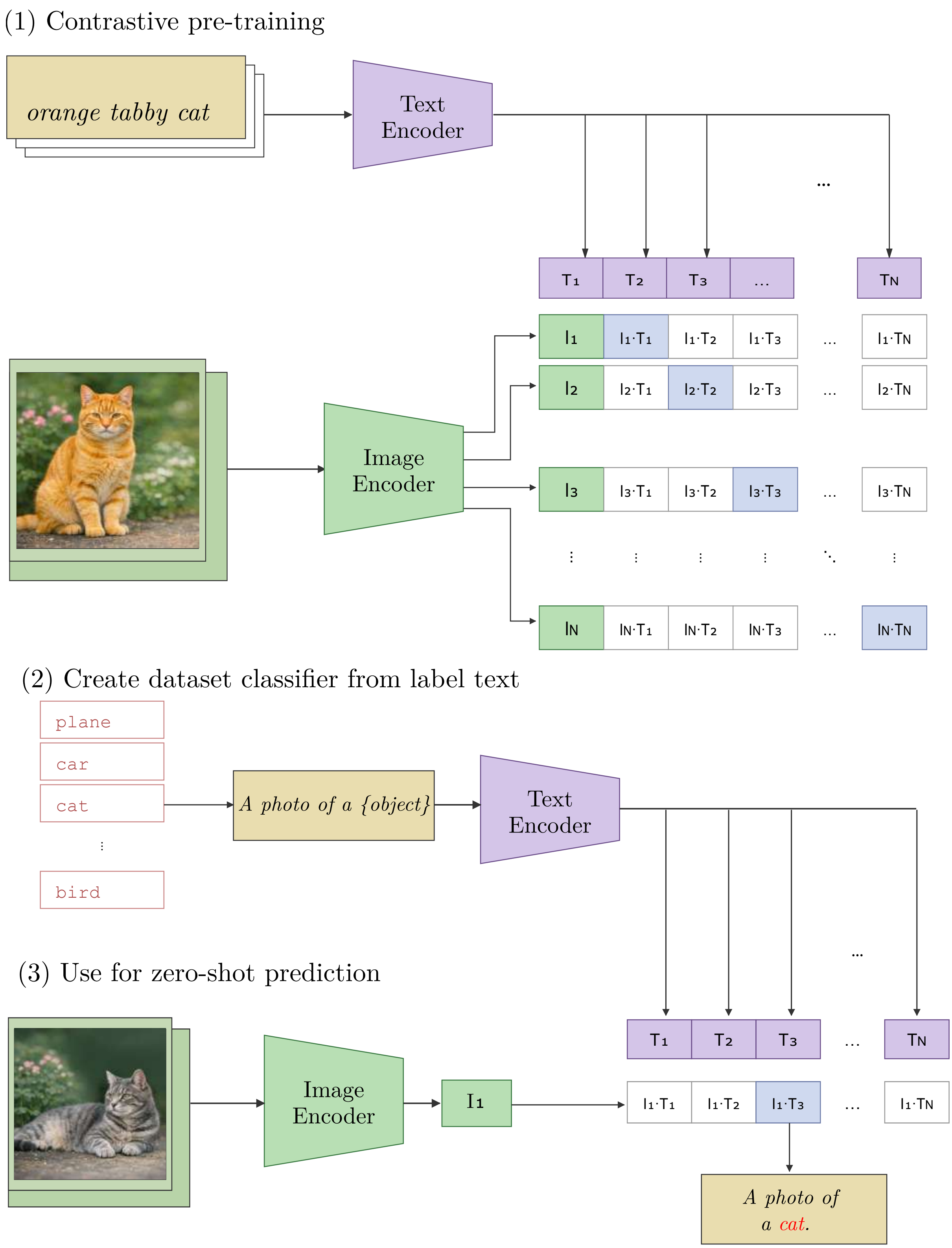

Figure 2.2: Overview of the CLIP training and zero-shot classification pipeline. Top: contrastive pre-training on image–text pairs. Bottom: zero-shot prediction by matching image and text embeddings [31].

2.3.2 Image Encoder

The image encoder in CLIP is implemented either as a convolutional neural network or as a Vision Transformer [31]. It processes the input image through multiple layers and produces a feature vector that is projected and normalized to lie on the unit hypersphere, enabling cosine similarity comparisons with text feature vectors.

2.3.3 Text Encoder

The text encoder is a Transformer-based language model that processes natural language descriptions into feature vectors [31]. An input description is tokenized into a sequence of tokens and passed through the Transformer, which applies self-attention across tokens to produce a single sentence-level representation. This vector is projected and normalized to the unit hypersphere, matching the dimensionality and scale of the image embeddings.

2.3.4 Contrastive Learning Objective

During pre-training, CLIP optimizes a contrastive objective that pulls matching image-text pairs together and pushes mismatched pairs apart in the shared embedding space. For a batch of N image-text pairs, both encoders produce normalized feature vectors and an $N \times N$ cosine similarity matrix is formed. A symmetric cross-entropy loss is applied over this matrix, treating the diagonal entries as correct pairs for both the image-to-text and text-to-image directions:

$$\mathcal{L} = -\frac{1}{2N} \sum_{i=1}^{N} \left[\log \frac{\exp(s_{ii}/\tau)}{\sum_{j=1}^{N} \exp(s_{ij}/\tau)} + \log \frac{\exp(s_{ii}/\tau)}{\sum_{j=1}^{N} \exp(s_{ji}/\tau)} \right] \tag{2.1}$$

where $s_{ij} = v_i^\top t_j$ is the cosine similarity between image embedding v_i and text embedding t_j, and τ is a learned temperature parameter that scales the similarity scores. This objective is optimized over hundreds of millions of image-text examples, enabling the model to learn broad visual and semantic concepts without manual annotation [31].

2.3.5 Zero-Shot Classification

Once trained, CLIP performs zero-shot classification by comparing the embedding of an input image against the embeddings of a set of textual class descriptions. Each class is described in natural language for example, *"a photo of a cat"* or *"a photo of a dog"* and encoded once by the text encoder to produce a normalized class embedding. An input image is encoded by the image encoder, and classification is performed by selecting the class description with the highest cosine similarity to the image embedding:

$$\hat{y} = \arg\max_{c} \frac{\exp(\cos(v, t_c)/\tau)}{\sum_{k=1}^{C} \exp(\cos(v, t_k)/\tau)} \tag{2.2}$$

where v is the normalized image embedding, t_c is the normalized text embedding for class c, C is the total number of classes, and τ is the temperature parameter. The softmax formulation converts raw similarity scores into a probability distribution over classes, and the predicted class is the one with the highest probability. This procedure requires no task-specific training data, as classification relies entirely on the learned image-text similarity from pre-training [31].

2.4 Geographic Information System (GIS)-Based Spatial Analysis

This section introduces the three core GIS operations underlying the proposed workflow: geo-referencing, buffer analysis, and zonal statistics, applied in sequence to extract spatial features from raster imagery.

2.4.1 Coordinate Reference Systems and Geo-referencing

Raster images are stored in pixel coordinates where each pixel position is given by its column and row indices. For spatial analysis, these pixel coordinates must be linked to a geographic coordinate reference system so that raster data can be combined with vector datasets such as points and polygons [52, 53]. This link is established through geo-referencing, which defines the correspondence between pixel positions and real-world coordinates. A common approach uses world files such as PNG world file (`.pgw`) that specify a six-parameter affine transformation from pixel coordinates to map coordinates [54]:

$$x_{\text{world}} = A \cdot x + B \cdot y + C \tag{2.3}$$

$$y_{\text{world}} = D \cdot x + E \cdot y + F \tag{2.4}$$

where A is the pixel width in map units, B and D are rotation terms equal to zero for north-up imagery, C and F are the real-world coordinates of the upper-left pixel centre, and E is the negative pixel height in map units [55].

2.4.2 Buffer Analysis

In geographic information systems, buffer analysis creates zones of a specified distance around geographic features such as points, lines, or polygons to study proximity relationships and neighbourhood effects [56, 57].These buffer zones are widely used to analyse potential impacts, accessibility, or the local context around objects of interest. For point features, a buffer is represented as a circular polygon with a user-defined radius r around the point location p, defined as the set of all locations whose distance to p does not exceed r:

$$\text{Buffer}(p, r) = \{\, x \mid \mathrm{d}(x, p) \leq r \,\} \tag{2.5}$$

When multiple concentric buffer zones are required, annular regions referred to as buffer rings are derived by subtracting the inner buffer from the outer buffer:

$$\text{Ring}(p, r_{\text{in}}, r_{\text{out}}) = \text{Buffer}(p, r_{\text{out}}) \setminus \text{Buffer}(p, r_{\text{in}}) \tag{2.6}$$

where r_{in} and r_{out} are the inner and outer radii respectively. Annular buffers isolate the spatial contribution of each distance band independently without including pixels from inner zones [56].

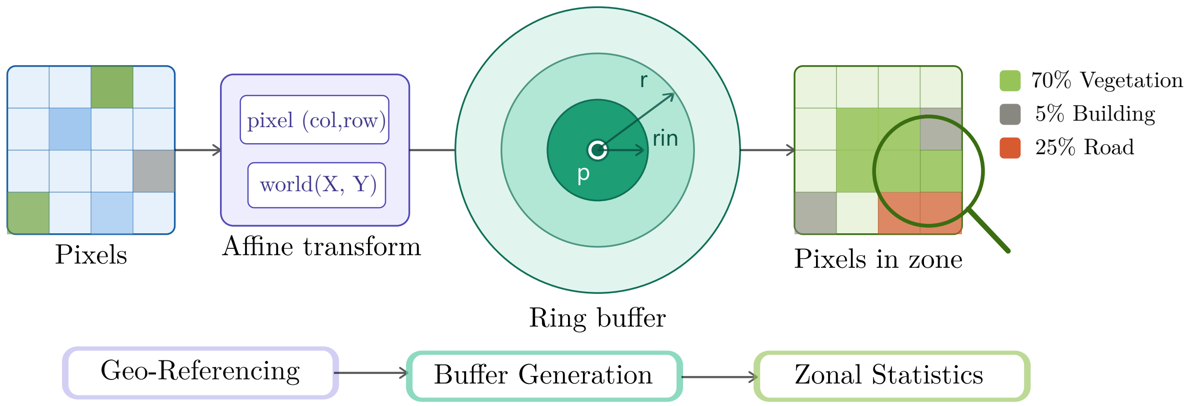

Figure 2.3: Schematic of the integrated spatial analysis workflow, showing the transformation from raw raster pixels to land cover composition estimates through geo-referencing, ring buffer generation, and zonal statistics.

2.4.3 Zonal Statistics

Zonal statistics summarize raster values within zones defined by spatial geometries such as polygons, buffers, or administrative units [58]. A zone is any spatial area sharing the same identifier, and statistics such as counts, sums, means, or proportions can be computed over all raster cells falling within each zone [59, 60]

Zonal statistics are widely used to relate gridded information to discrete spatial units. Examples include summarizing land-cover, elevation, or environmental indicators at the level of parcels or landscape patches [60]. When applied to categorical raster data, zonal statistics quantify the proportion of pixels belonging to each category within a zone, providing per-class coverage estimates across spatial units of interest.

For example, consider a categorical land cover raster with three classes: Vegetation, Building, and Road. Applying zonal statistics within a circular buffer around a point location would compute the percentage of pixels belonging to each class inside that buffer, yielding a composition vector such as [70% Vegetation, 5% Building, 25% Road]. This determines the surrounding land use composition for any spatial location of interest.

3 METHODOLOGY

This chapter describes the design and implementation of a zero-shot urban land use classification pipeline that requires no task-specific labelled training data, as illustrated in Figure 3.1. In the first stage, SAM was applied to each aerial image patch to partition it into a set of spatially coherent regions without semantic labels (Section 3.2). In the second stage, CLIP assigned a land use class label to each segmented region using manually designed aerial text prompts, without fine-tuning on labelled aerial imagery (Section 3.3). In the third stage, concentric annular buffer zones at radii of 2.5 m, 5.0 m, and 7.5 m were generated around each tree location, and the proportion of each land use class within each zone was quantified against pixel-level ground truth (Section 3.4).

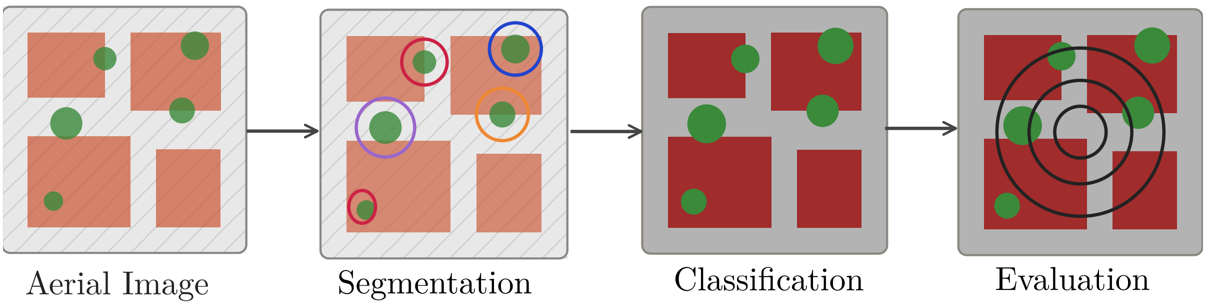

Figure 3.1: Zero-shot urban land use classification pipeline overview.

All segmentation and classification experiments were executed on the National High Performance Computing Centre (NHR@FAU) at Friedrich-Alexander-Universität Erlangen-Nürnberg using GPU-enabled compute nodes. Evaluation was performed on CPU without GPU acceleration.

3.1 Dataset

3.1.1 Study Area

The study area was the city of Erlangen, located in Bavaria, Germany. The city contains a diverse mix of residential areas, commercial zones, public green spaces, and road infrastructure, representing the range of urban land cover types targeted by the classification pipeline. Aerial imagery and IoT-based tree monitoring data were available for Erlangen, making it a suitable study site for the proposed evaluation framework.

3.1.2 Aerial Imagery

Three aerial image sources were used in this study: one for pipeline development and optimisation (Aerial 2020), and two for cross-dataset generalisation evaluation (Aerial 2025 and Google Satellite). All three sources covered the same 95 tree locations, enabling direct performance comparison across temporal, seasonal, and sensor variations. Figure 3.2 shows the spatial distribution of the 95 tree locations across the Erlangen study area.

Figure 3.2: Spatial distribution of the 95 monitored street tree locations across Erlangen, Germany, overlaid on the 2020 aerial orthophoto. Red dots indicate individual tree positions used for pipeline development and cross-dataset evaluation.

Aerial 2020 (Development Dataset)

High-resolution aerial images were obtained from the City of Erlangen's publicly available orthophoto dataset (Luftbild 2020), acquired on 28 March 2020. The dataset covers approximately 140 km^2 of the Erlangen city area at a native ground sampling distance (GSD) of 20 cm/pixel in the RGB colour space. All images were geo-referenced in the ETRS89/UTM Zone 32N coordinate reference system (EPSG:25832). Images were captured in early spring, depicting trees in a leafless state prior to the growing season. The dataset is licensed under Creative Commons Attribution 4.0 International (CC BY 4.0; © *Stadt Erlangen*) [61].

This dataset served as the basis for all pipeline development, including SAM parameter tuning (Section 3.2), CLIP prompt optimisation (Section 3.3), and initial performance evaluation (Section 3.4).

Aerial 2025 (Generalisation Dataset)

Aerial images from summer 2025 were obtained from the Bavarian state aerial survey (Bayernbefliegung), which covers Mittelfranken on a 2-year rotation cycle during summer months (typically April–August) [62]. The imagery depicts trees with full foliage. Ground sampling distance is approximately 20 cm/pixel, and the same coordinate reference system (EPSG:25832) was used for geo-referencing. No pipeline modifications, retraining, or parameter adjustments were performed for this dataset.

Google Satellite (Generalisation Dataset)

Google Satellite imagery was accessed via the Google Earth tile service [63]. The imagery consists of multi-temporal composites typically from summer periods with full tree foliage. Native tiles were provided in EPSG:3857 and reprojected to EPSG:25832 for consistency. Spatial resolution varies but is estimated at 15–30 cm ground sampling distance. As with Aerial 2025, the pipeline configuration was applied without modification.

3.1.3 Tree Inventory and IoT Sensor Dataset

Tree locations were sourced from the Agvolution Climavi City system[1], an Internet of Things (IoT)-based urban tree monitoring platform that deploys soil sensors at individual tree locations to measure environmental parameters such as soil moisture and temperature. The broader dataset comprises 386 sensor-equipped trees across 33 cities in Central Europe. The Erlangen subset was used exclusively in this study, comprising trees from two deployments: the City of Erlangen municipal project and the University of Erlangen Green Office project, yielding 95 aerial image patches across 95 unique tree locations. Each tree is uniquely identified by a device Extended Unique Identifier (devEUI) assigned during IoT sensor deployment. Tree location coordinates (EPSG:25832) and associated metadata were managed within QGIS vector layers (`treeLocations.shp`), serving as the spatial reference for all image extraction and annotation steps.

3.1.4 Image Patch Extraction

For each of the 95 tree locations, aerial image patches were extracted from the Luftbild 2020 orthophoto using an automated QGIS Python console script. Each patch was rendered as a 1024 × 1024 pixel image at a fixed map scale of 1:500, centred on the tree's GPS coordinates. This resolution matches the native input size of the SAM image encoder, which rescales all input images to 1024 × 1024 pixels prior to inference [30], avoiding resampling artefacts in the segmentation step. Three resolution variants were generated per location by varying the output dots per inch (DPI) at constant pixel dimensions, as summarised in Table 3.1.

Figure 3.3 (a) to (c) illustrates the effect of DPI on ground coverage and object-level detail for a representative tree location. At fixed pixel dimensions, the DPI setting determined

[1]Dataset currently under review for publication. Citation to be added upon publication.

Level	DPI	Ground coverage (m)	GSD (cm/px)	File suffix
Fine	300	43.35×43.35	4.23	_1
Medium	225	57.80×57.80	5.64	_2
Coarse	150	86.70×86.70	8.47	_3

Table 3.1: Multi-resolution image export configuration at fixed map scale 1:500 and fixed output size 1024×1024 px.

the ground area compressed into the 1024×1024 pixel grid. Higher DPI produced finer object details over a smaller ground area. Lower DPI covered a larger ground area at coarser detail. The SAM segmentation model processes all variants identically as 1024×1024 pixel arrays, as DPI metadata is not used during inference.

Each exported image was accompanied by a PGW world file encoding six affine transformation parameters: pixel size in the X and Y directions, two rotation terms (zero for north-up imagery), and the real-world coordinates of the upper-left pixel centre. These parameters enabled pixel-to-coordinate conversion and were used during multi-scale mask fusion to spatially align segmentation outputs across resolution levels (Section 3.2.4).

Images were named following the convention `{devEUI}_{suffix}.png` with corresponding world files `{devEUI}_{suffix}.pgw`. The 225 DPI variant (_2) served as the primary scale throughout classification and evaluation, providing a balanced trade-off between ground coverage and object-level detail. The total dataset comprised 285 images across 95 tree locations, with three resolution variants per location at 300, 225, and 150 DPI, each accompanied by a corresponding `.pgw` world file.

(a) 300 DPI (b) 225 DPI (c) 150 DPI (d) Ground Truth

Figure 3.3: Representative aerial image patches for a single tree location. Panels (a) to (c) shows the same location at three resolution levels: (a) 300 DPI (fine), (b) 225 DPI (medium, reference), and (c) 150 DPI (coarse). Panel (d) shows the pixel-level ground truth overlay for the Vegetation class (green). The red cross marks the tree centre point, and evaluation rings at 2.5 m, 5.0 m, and 7.5 m are shown as dotted yellow circles.

3.1.5 Ground Truth Annotation

Land use polygons were manually annotated in QGIS by four contributors across dedicated annotation sessions. Each annotator followed a shared protocol developed for this project. The workflow proceeded as follows: the tree location was identified in QGIS using the devEUI from the IoT sensor layer, the corresponding image layer was selected for editing, and vegetation and building polygons were digitised manually using the polygon and curve tools within each buffer radius. Annotations were saved per layer (greenAtta, greenDeta, buildings) and linked to the tree location via devEUI. No formal inter-annotator agreement assessment was performed, as the annotation targets (vegetation patches and building footprints visible in high-resolution aerial imagery) were considered sufficiently unambiguous for the urban monitoring context of this study. The annotations recorded vegetation, building, and sealed surface coverage within buffer radii of 2.5 m, 5.0 m, and 7.5 m around each tree point. Pixel-level ground truth masks were derived from these vector layer annotations, stored in six shapefiles per buffer radius `greenAtta{r}.shp`, `greenDeta{r}.shp`, and `buildings{r}.shp`, where `{r}` denoted the buffer radius. All shapefiles were projected to ETRS89/UTM Zone 32N (EPSG:25832). Three land use classes were defined:

- **Vegetation**: pixels covered by attached green space (`greenAtta`) or detached green space (`greenDeta`) polygons.
- **Building**: pixels covered by building footprint polygons.
- **Road**: the remaining class, comprising all pixels not labelled as Vegetation or Building, including roads, pavements, sealed surfaces, and bare ground. This residual definition groups visually distinct surface types under a single label, making the class inherently heterogeneous. This choice was made to reduce annotation effort across 95 tree locations and reflects the application objective, where the primary distinction of interest is between vegetated and non-vegetated surfaces rather than between individual sealed surface types. The resulting heterogeneity is acknowledged as a factor contributing to higher classification difficulty for this class.

Vegetation and Building masks were rasterized independently for each buffer radius using `rasterio.features.rasterize` and accumulated across all three radii through bitwise OR, producing one full-image binary mask per class. The Road mask was derived as the pixel-wise residual:

$$\text{Road} = \neg\,(\text{Vegetation} \vee \text{Building}) \tag{3.1}$$

Figure 3.3(d) shows the pixel-level ground truth overlay with evaluation rings for the same tree location. The three binary masks were saved as compressed `.npz` files named `{devEUI}_2_gt.npz`. Ring-level separation into annular evaluation zones was performed during evaluation rather than during ground truth generation (Section 3.4.2).

Figure 3.4 shows the ground truth land use distribution per class and buffer ring across the 95 tree locations. Vegetation coverage dominated Ring 1 and decreased with increasing buffer distance, while Building coverage remained consistently low across all rings, resulting

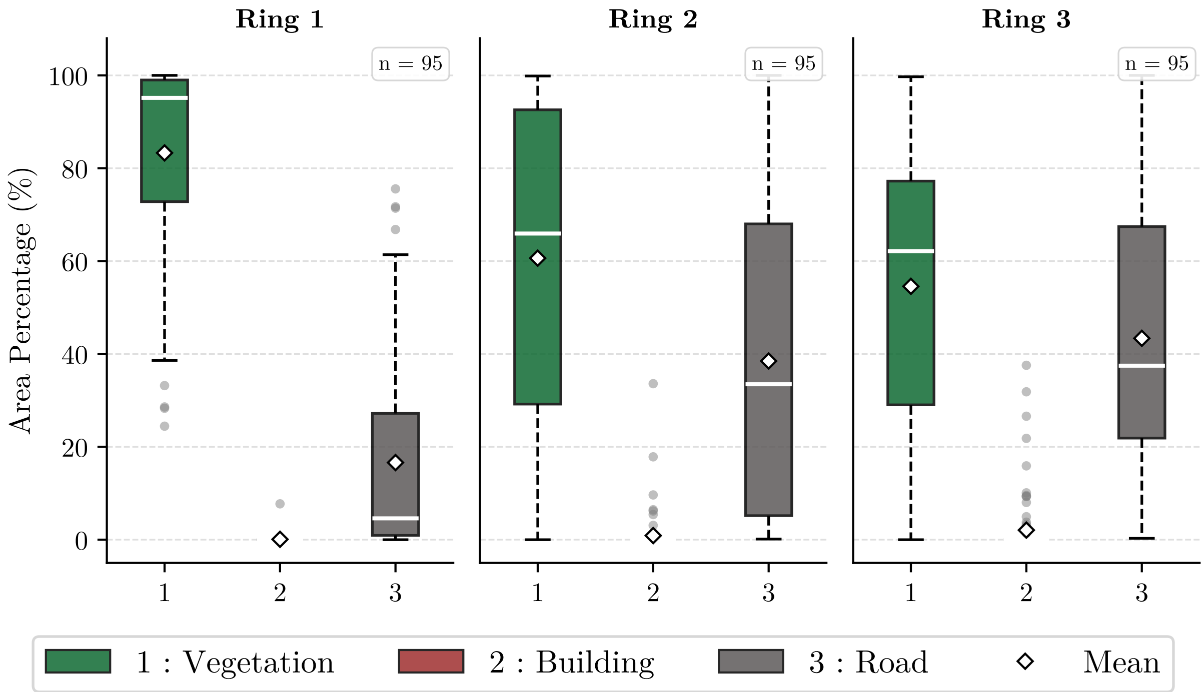

Figure 3.4: Ground truth land use distribution per buffer ring across 95 Erlangen tree locations. Boxes show the interquartile range, with median lines, mean diamonds, whiskers extend to 1.5 × IQR, and outliers as grey circles.

in a notable class imbalance that is considered when interpreting classification performance in Chapter 4.

3.2 Zero-Shot Aerial Image Segmentation with SAM

3.2.1 SAM Model

Three variants of SAM were evaluated on a 15-image subset at 225 DPI to select an appropriate model capacity: ViT-Base (ViT-B, 91M parameters), ViT-Large (ViT-L, 308M parameters), and ViT-Huge (ViT-H, 636M parameters) [30]. The subset size was chosen to limit computational cost during the initial backbone comparison. Each variant was evaluated under default parameter configurations (Table 3.2) to assess model capacity under consistent conditions. The selected backbone (ViT-H) was subsequently applied to the full 95-image dataset across all six segmentation experiments, where its coverage advantage was confirmed at scale. SAM was operated in automatic mask generation mode, in which a dense grid of point prompts was systematically sampled across the image and candidate masks were generated without user interaction. The implementation used the official SAM package (version 1.0, Meta AI Research) with PyTorch 2.0.1 and CUDA 11.8.

3.2.2 Parameter Configurations

Two parameter configurations were evaluated: a default setup following the SAM authors' recommended settings and a fine-tuned setup determined through iterative testing on the dataset. The parameter values are listed in Table 3.2.

Parameter	Default	Fine-tuned
`points_per_side`	32 (1,024 points)	64 (4,096 points)
`pred_iou_thresh`	0.88	0.80
`stability_score_thresh`	0.95	0.85
`min_mask_region_area`	0 px	100 px

Table 3.2: SAM automatic mask generation parameter configurations. Default values follow [30]. Fine-tuned values were determined through iterative testing on the dataset.

Increasing `points_per_side` from 32 to 64 raised the number of prompt points from 1,024 to 4,096, targeting improved detection of small objects at the cost of increased processing time. Lowering the IoU and stability thresholds increased mask detection sensitivity by accepting masks with lower predicted quality scores. Setting `min_mask_region_area` to 100 pixels filtered out small fragmented regions that may be introduced by the lower quality thresholds.

3.2.3 Single-Scale Segmentation Procedure

For single-scale experiments, segmentation was performed on each 225 DPI image using the following procedure:

1. The input image was loaded into GPU memory.
2. SAM generated candidate masks using the configured automatic mask generation parameters.
3. Masks were filtered by the `pred_iou_thresh` and `stability_score_thresh` thresholds.
4. Non-maximum suppression (NMS) removed redundant overlapping masks.
5. Masks with area below `min_mask_region_area` were discarded.
6. Remaining masks were stored as binary spatial arrays in compressed `.npz` format using `np.savez_compressed`, together with per-mask metadata comprising bounding box, area, predicted IoU, stability score, and perimeter.

3.2.4 Multi-Scale Segmentation and Fusion

For multi-scale experiments, steps 1–6 of Section 3.2.3 were applied independently to all three resolution variants (300, 225, and 150 DPI) of each tree location. The 225 DPI variant served as the reference coordinate system as it provided a balanced trade-off between ground coverage and object-level detail (Section 3.1.4). Masks from the 300 DPI and 150 DPI images were reprojected into the 225 DPI coordinate system using affine transformations derived from the PGW world files associated with each image. Reprojected masks with an area below 50 pixels were discarded to remove transformation artefacts. The complete fusion procedure is illustrated in Figure 3.5.

Pairwise Intersection-over-Union was computed for all mask pairs across resolution levels as defined in Equation 3.2. Masks with IoU ≥ 0.60 were identified as representing the same object and merged through connected component analysis, with per-mask metadata recomputed as area-weighted averages over the merged group. The threshold of 0.60 was selected through empirical testing at three candidate values: a threshold of 0.30 merged spatially adjacent but distinct objects, while a threshold of 0.99 missed duplicate detections of the same object across resolution levels. The three candidate values were chosen to represent qualitatively distinct merging behaviours: aggressive merging (0.30), conservative merging (0.99), and a balanced intermediate (0.60). A finer grid search was not performed, as the qualitative failure modes observed at 0.30 and 0.99 indicated that 0.60 was a reasonable operating point rather than a boundary case requiring precise calibration. A threshold of 0.60 avoided both failure modes and was therefore adopted for all multi-scale experiments.

$$\mathrm{IoU}(m_i, m_j) = \frac{|m_i \cap m_j|}{|m_i \cup m_j|} \tag{3.2}$$

3.2.5 Proxy Evaluation Metrics

These unsupervised metrics evaluate segmentation quality in the absence of pixel-level ground truth.

Pixel Coverage Ratio (C, %):

$$C = \frac{\text{Total pixels covered by any mask}}{\text{Total image pixels}} \times 100 \tag{3.3}$$

C measures the proportion of the image area assigned to at least one mask and served as the primary comparison metric across all configurations.

Median Mask Area (A_{med}, pixels):

$$A_{\text{med}} = \text{median}\left(\{\text{Area}(M_i)\}_{i=1}^{N}\right) \tag{3.4}$$

where N is the total number of masks and M_i denotes mask i. A_{med} characterises the typical spatial extent of individual segments, where low values indicate fragmented output and high values indicate larger and more coherent regions.

Additionally, the following metrics were recorded per experiment: segment count (total number of masks generated per image), mean predicted IoU (SAM's internal mask quality estimate, computed independently of ground truth), and processing time (seconds per image).

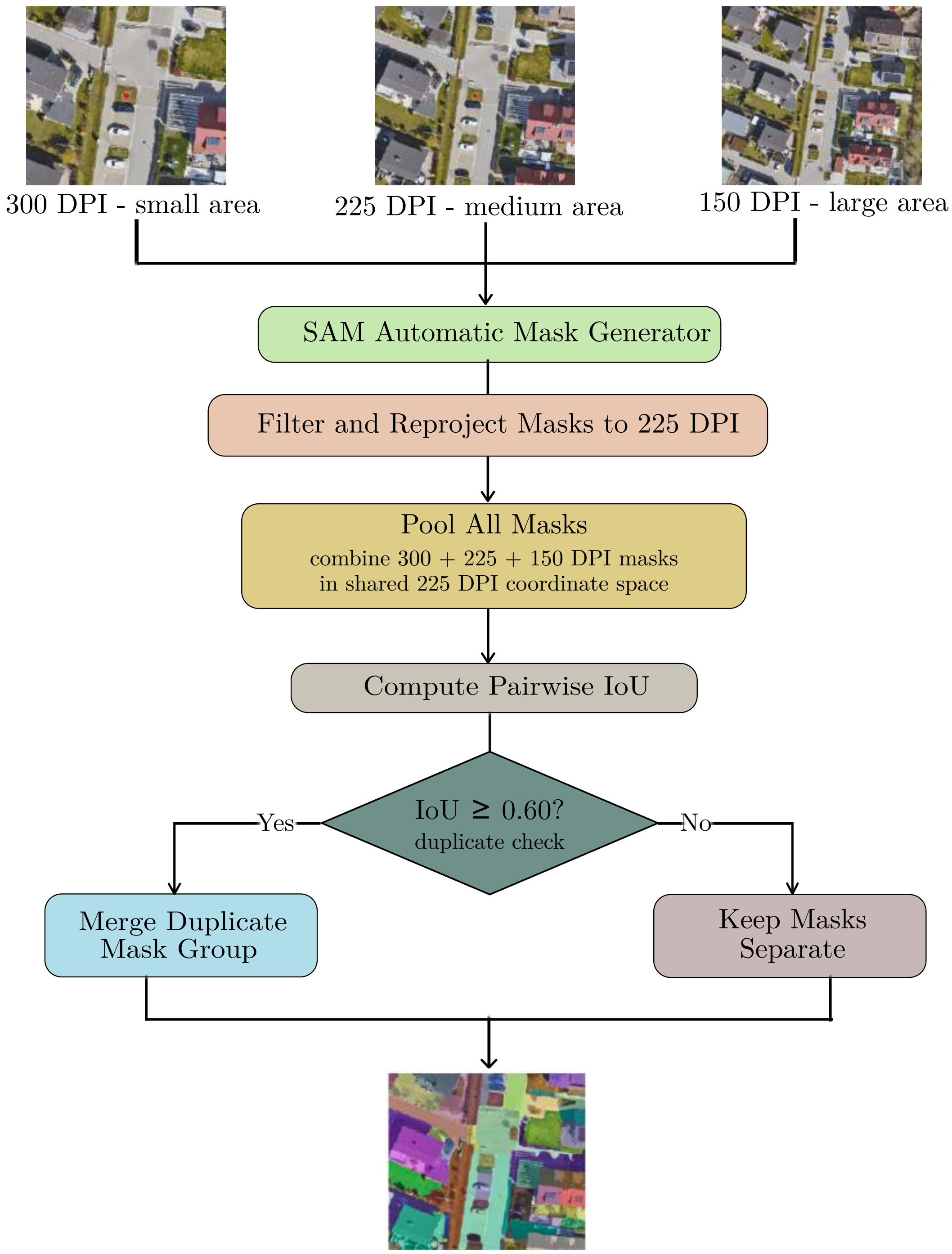

Figure 3.5: SAM multi-scale segmentation and fusion procedure comprising three parallel resolution streams, pairwise IoU-based duplicate detection, and mask merging into a consolidated 225 DPI output.

3.2.6 Experimental Configurations

Six configurations were evaluated to assess the effect of parameter tuning and resolution combinations on segmentation quality (Table 3.3). E1 established the single-scale baseline. E2 isolated the effect of parameter tuning at 225 DPI. E3 and E4 evaluated multi-scale fusion under default and fine-tuned parameters, respectively. E5 and E6 tested the contribution of individual resolution levels through ablation: E5 excluded the 300 DPI scale and E6 excluded the 150 DPI scale. All experiments were conducted on the full 95-image dataset. Metrics are reported as mean ± standard deviation across all 95 tree locations.

ID	**Resolution(s)**	**Parameters**	**Purpose**
E1	225 DPI	Default	Single-scale baseline
E2	225 DPI	Fine-tuned	Parameter optimisation
E3	300 + 225 + 150 DPI	Default	Multi-scale baseline
E4	300 + 225 + 150 DPI	Fine-tuned	Combined optimisation
E5	225 + 150 DPI	Fine-tuned	Ablation: exclude 300 DPI
E6	300 + 225 DPI	Fine-tuned	Ablation: exclude 150 DPI

Table 3.3: SAM segmentation experimental configurations. All multi-scale experiments used a fusion IoU threshold of 0.60 and a minimum post re-projection mask area of 50 pixels.

3.3 Zero-Shot Classification with CLIP

3.3.1 CLIP Model

The CLIP model was used for zero-shot semantic classification of SAM segments. The ViT-Large/patch14-336 variant was loaded from a local checkpoint via the HuggingFace Transformers API [31, 64]. The model accepted segmented masks resized to 336×336 pixels.

CLIP computes similarity between image and text embeddings through contrastive alignment trained on 400 million image-text pairs. Classification was performed without fine-tuning on labelled aerial imagery, constituting a zero-shot approach similar to recent CLIP-based methods in remote sensing [65, 66]. Three land use classes were defined: Vegetation, Building, and Road, following the class definitions in Section 3.1.5.

3.3.2 Classification Input

SAM segmentation masks from the selected segmentation configuration served as input. For each tree location, binary segment masks in 225 DPI reference coordinates were loaded from `.npz` files alongside the corresponding aerial image patch.

3.3.3 Text Prompts

Text prompts described the visual appearance of each land use class from an aerial perspective. Two prompt categories were evaluated: a single-prompt baseline following the standard CLIP zero-shot template [31], and multi-prompt aerial descriptors designed specifically for overhead imagery. Designing domain-specific prompts has been shown to improve zero-shot performance of CLIP-style models in remote sensing and aerial scene classification [65, 66].

Single-Prompt Baseline (Prompt 0)

Following the standard zero-shot classification template from [31], one generic prompt per class was used (Table 3.4). This configuration established a domain-mismatch lower bound, as the template was designed for ground-level natural images rather than aerial imagery.

Class	**Prompt**
Vegetation	*"a photo of vegetation"*
Road	*"a photo of a road"*
Building	*"a photo of a building"*

Table 3.4: Single-prompt baseline (Prompt 0) following the standard CLIP zero-shot template [31].

Multi-Prompt Aerial Descriptors (Prompts 1-1.3)

Multi-prompt versions used three to four descriptions per class. Prompts were designed using a combination of colour, material, and geometric language suited to overhead aerial imagery. Four versions were developed iteratively through the experimental process described in Section 3.3.7. The final configuration (Prompt 1.3) is shown in Table 3.5.

The evolution of prompt versions across the experimental process is documented in Table 3.7 (Section 3.3.7).

3.3.4 Segment Crop Preparation

Each binary segment mask was converted into a 336×336 pixel image crop for CLIP inference. Four crop-preparation strategies were designed and evaluated, motivated by the importance of combining local object appearance with surrounding context in vision models [67, 68], and by recent efforts to couple SAM's spatial precision with CLIP's semantic priors [44, 69, 70].

Zero (context-free baseline). Non-mask pixels were set to zero (black). A tight crop around the mask bounding box was extracted and resized to 336×336 pixels with

Class	Prompts
Vegetation	*"green trees or bushes seen from above"*; *"aerial view of a grassy lawn or field"*; *"leafy tree canopy from above"*; *"natural vegetation in green or brown from above"*
Road	*"long narrow strip of asphalt for cars from above"*; *"grey road or street with lane markings from above"*; *"aerial view of a paved road surface"*; *"dark gray pavement for vehicles from above"*
Building	*"red tile rooftop made of clay from above"*; *"gray flat rooftop made of concrete from above"*; *"building roof with shingles or tiles from above"*

Table 3.5: Multi-prompt aerial descriptors, final configuration (Prompt 1.3). Each class is described by three to four colour and material-based prompts designed for overhead imagery.

aspect-ratio-preserving padding. CLIP received only the isolated segment content with no surrounding information.

Highlight. A semi-transparent cyan overlay (RGB $(0, 255, 255)$ at opacity $\alpha = 0.35$) was applied to the target segment on the full image. A crop three times the bounding box size, centred on the segment centroid, was extracted. CLIP received surrounding real pixels with a visual cue marking the target segment. The cyan overlay colour (RGB $(0, 255, 255)$) was selected for its high contrast against the typical red, green, and brown tones of urban aerial imagery, ensuring the target segment remained visually distinguishable. The opacity of $\alpha = 0.35$ was chosen to preserve the underlying image texture within the segment while still providing a clear visual indicator of the region of interest. Both values were determined through visual inspection rather than systematic optimisation.

Larger Crop. A region three times the segment bounding box, centred on the segment centroid, was extracted without any overlay or background replacement. CLIP received genuine surrounding pixels with no explicit segment indicator. The factor of three times the bounding box size was chosen empirically to provide sufficient surrounding context while keeping the target segment visually prominent within the CLIP input. Smaller factors risked excluding relevant neighbouring features, while larger factors reduced the relative size of the target segment within the crop. This value was determined through visual inspection rather than systematic optimisation and represents a reasonable default rather than a precisely calibrated parameter.

Dual Composite. A 336×336 side-by-side composite was constructed: the left half (168×336 pixels) showed the full image with cyan highlight on the target segment and the right half showed the isolated segment on a black background. CLIP received both global context and local segment detail in a single forward pass.

All crops were resized to 336×336 pixels using aspect-ratio-preserving padding. The four strategies are illustrated in Figure 3.6.

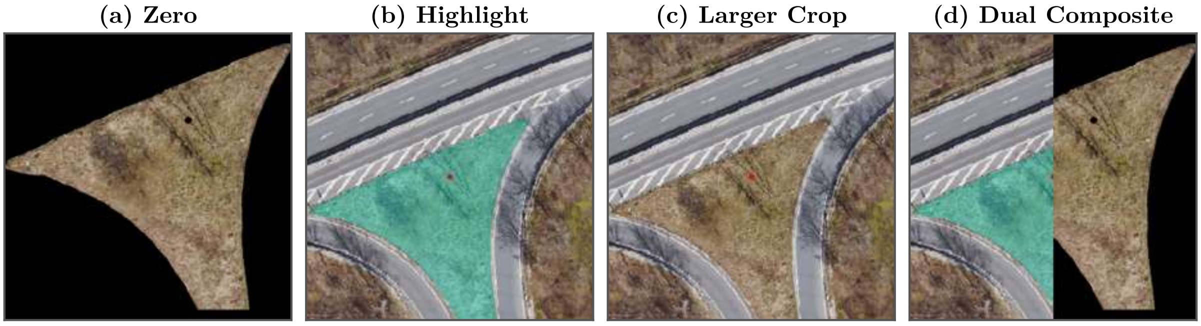

Figure 3.6: Four segment crop-preparation strategies for CLIP inference. (a) Zero: isolated segment on black background. (b) Highlight: cyan overlay marking the target segment within a three-times enlarged crop. (c) Larger Crop: three-times enlarged region with no segment indicator. (d) Dual Composite: context with highlight (left) and isolated segment (right) combined in a single 336×336 pixel input.

3.3.5 CLIP Inference and Classification

For each segment crop, CLIP computed softmax probabilities over all text prompts jointly, using cosine similarity with a learned temperature parameter as in [31]:

$$p_{ij} = \frac{\exp\Big(\cos(v_i,\, t_j) \,/\, \tau\Big)}{\sum_{k=1}^{K} \exp\Big(\cos(v_i,\, t_k) \,/\, \tau\Big)}. \tag{3.5}$$

where v_i denoted the image embedding of crop i, t_j denoted the text embedding of prompt j, τ was the learned temperature parameter, and K was the total number of prompts across all classes.

Two aggregation modes were evaluated to combine scores from multiple prompts.

Single aggregation (argmax over all prompts).

$$\hat{y}_i = \text{class}\left(\arg\max_j \; p_{ij}\right), \qquad c_i = \max_j \; p_{ij} \tag{3.6}$$

The segment was assigned to the class of the single highest-scoring prompt. Confidence was the corresponding softmax probability.

Average aggregation (mean probability per class).

$$\bar{p}_{ic} = \frac{1}{|P_c|} \sum_{j \in P_c} p_{ij}, \qquad \hat{y}_i = \arg\max_c \ \bar{p}_{ic} \tag{3.7}$$

where P_c denoted the set of prompt indices belonging to class c. Confidence was the winning class-average probability.

Inference was performed in batches of 16 crops. Classification results were saved per tree location as `.npy` files containing predicted class labels and confidence scores for all segments.

3.3.6 Proxy Evaluation Metrics

The following intrinsic metrics were computed per image to monitor classification quality in the absence of ground truth. They are simple, task-specific measures inspired by work on estimating segmentation quality without labels and enforcing spatial consistency between neighbouring predictions [71, 72].

Mean Confidence.

$$\bar{c} = \frac{1}{N} \sum_{i=1}^{N} c_i \tag{3.8}$$

where N was the total number of segments and c_i was the confidence of segment i. Higher values indicated more decisive predictions.

Low-Confidence Percentage. Proportion of segments with confidence below 0.30. The threshold of 0.30 was chosen as a conservative lower bound on CLIP softmax confidence, below which predictions were considered unreliable. Since CLIP distributes probability across all prompts, a segment classified with confidence below 0.30 indicates no prompt achieved a clearly dominant score. This value was selected through empirical inspection of the confidence distributions observed during initial test runs rather than from a systematic threshold search.

Spatial Consistency Score.

$$\mathrm{SC} = \frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} \frac{\left|\{j \in \mathcal{N}_i : \hat{y}_j = \hat{y}_i\}\right|}{|\mathcal{N}_i|} \tag{3.9}$$

where $\mathcal{N}_i$ was the set of segments whose centroids fell within 50 pixels of segment i, and $\mathcal{A}$ was the set of non-isolated segments. The neighbourhood radius of 50 pixels was chosen to capture immediately adjacent segments at the 225 DPI reference resolution, where 50 pixels corresponds to approximately 2.8 m on the ground. This distance was considered appropriate for capturing local land use context around each segment without extending into distant scene regions. The value was determined through visual inspection of typical segment sizes and spatial arrangements in the dataset rather than through systematic optimisation.

3.3.7 Experimental Design

Classification was optimised through five sequential experiments, each motivated by findings from the previous step. This staged design mirrors recent work that incrementally adapts foundation models such as CLIP and SAM to downstream segmentation tasks [44, 69, 70]. The complete experimental sequence is summarised in Table 3.6.

Step	Approach	Prompts	Aggregation	Purpose
1	zero	Prompt 0	single	Establish baseline lower bound
2	zero	Prompt 1	single	Test aerial multi-prompt design
3	all four	Prompt 1	single	Select best context approach
4	zero	Prompt 1 - 1.3	single	optimise prompt version
5	zero	Prompt 1.3	single vs. average	Select aggregation mode

Table 3.6: CLIP classification experimental sequence. Each step was motivated by results from the previous step. Buffer eval. refers to evaluation against pixel-level ground truth (Section 3.4)

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Step 1: Baseline Experiment

The zero crop-preparation strategy was combined with the standard CLIP single-prompt baseline (Prompt 0) on the complete 95-tree dataset. This established a domain-mismatch lower bound by applying ground-level image templates to aerial imagery. Proxy metrics and GIS buffer evaluation results from this configuration served as the lower bound reference for subsequent prompt design steps.

Step 2: Initial Aerial Prompt Design

Prompt 0 was replaced with Prompt 1, a multi-prompt aerial descriptor set designed using colour and material language. The zero approach was retained to isolate the effect of prompt design. Results were evaluated using proxy metrics and GIS buffer ground truth and compared against Step 1 to quantify the effect from aerial-specific multi-prompt design over the generic CLIP baseline.

Step 3: Context Approach Comparison

With Prompt 1 evaluated against Prompt 0, all four crop-preparation strategies were evaluated using Prompt 1 and single aggregation on the complete 95-tree dataset. Each approach was first assessed using proxy metrics and qualitative visual inspection of classification overlay maps. Since proxy metrics and visual inspection did not produce a consistent ranking across approaches, all four approaches were additionally evaluated against GIS buffer ground truth (Section 3.4). The approach yielding the best agreement with ground truth was selected for subsequent steps.

Step 4: Prompt Optimisation

With the context approach selected in Step 3, four multi-prompt versions were evaluated sequentially (Table 3.7). Each version was tested on the complete 95-tree dataset and evaluated against GIS buffer ground truth. The version yielding the best ground truth agreement was selected as the final prompt configuration.

Version	Change from previous	Motivation
Prompt 1	Multi-prompt aerial descriptors	Address domain gap with added material and colour detail
Prompt 1.1	Building prompts revised	Geometry-based prompts caused rooftop/road confusion replaced with material descriptors
Prompt 1.2	Vegetation prompts expanded	Added brown/yellow descriptor to handle dried or seasonal vegetation
Prompt 1.3	Road and Building prompts expanded	Added one Road prompt for emphasis balanced prompt counts across classes

Table 3.7: Prompt version history. Each version introduced a targeted change motivated by classification failures observed in the previous version.

Step 5: Aggregation Mode Selection

With the context approach and prompt version fixed, single and average aggregation modes were compared on the complete Erlangen dataset using both proxy metrics and GIS buffer ground truth evaluation. The aggregation mode producing more consistent results across both evaluation frameworks was selected as the final classification configuration and applied to all tree locations for downstream buffer analysis.

3.4 GIS Buffer Analysis and Evaluation

3.4.1 Overview

Classification performance was evaluated by comparing per-pixel CLIP predictions against pixel-level ground truth masks within three concentric annular buffer zones around each tree location. Evaluation was performed on the complete 95-tree dataset for each experimental configuration described in Section 3.3.7. All evaluation steps were executed on CPU.

3.4.2 Annular Buffer Ring Definition

Three concentric annular buffer rings were defined around each tree location to assess land use composition at increasing distances from the tree (Table 3.8). Annuli rather than full circles were used so that each ring captured only the land use at its specific distance band, without including pixels from inner zones.

Ring	r_{in} (m)	r_{out} (m)	Area (m²)	Pixels
Ring 1	0.0	2.5	19.6	6,162
Ring 2	2.5	5.0	58.9	18,488
Ring 3	5.0	7.5	98.2	30,814

Table 3.8: Buffer ring definitions. Theoretical areas are computed as $\pi r_{\text{out}}^2 - \pi r_{\text{in}}^2$. Pixel counts are approximate and correspond to the 225 DPI image grid (0.0564 m/pixel).

Ring masks were computed at runtime for the 225 DPI image grid (1024 × 1024 pixels) using Euclidean distance from the image centre $(r_0, c_0) = (512, 512)$:

$$d_{rc} = \sqrt{(c - c_0)^2 + (r - r_0)^2} \tag{3.10}$$

A pixel at position (r, c) was assigned to ring k if:

$$\frac{r_{\text{in},k}}{s} < d_{rc} \leq \frac{r_{\text{out},k}}{s} \tag{3.11}$$

where $s = 0.0564$ m/pixel was the ground sampling distance of the 225 DPI image, computed as:

$$s = \frac{1}{DPI} \times 0.0254 \times \text{SCALE} \tag{3.12}$$

with DPI = 225 and SCALE = 500.

3.4.3 Prediction Layer Construction

CLIP classification results (per-segment class labels) were combined with SAM segmentation masks to produce a per-pixel prediction layer of shape 1024×1024. For each segment mask M_i with predicted class label $\hat{y}_i$, the corresponding class index was assigned to all pixels covered by that mask:

$$L[r, c] = \mathrm{idx}(\hat{y}_i) \quad \forall\, (r, c) \in M_i \tag{3.13}$$

where class indices were: Vegetation = 1, Building = 2, Road = 3. Pixels not covered by any SAM segment retained index = 0 in the prediction layer and were excluded from all evaluation metrics. Only pixels assigned a class label by CLIP contributed to coverage percentages and binary overlap computations. Evaluation was performed exclusively within the three annular buffer rings (Section 3.4.2), and pixels outside the buffer zones were not considered regardless of their predicted or ground truth class. Ground truth Road retained its residual definition comprising all pixels not labelled as Vegetation or Building within the buffer zones, meaning predicted Road and ground truth Road are not symmetrically defined. This asymmetry may result in systematically lower Road recall, as unsegmented pixels within the buffer zones contribute to ground truth Road but not to predicted Road.

3.4.4 Evaluation Metrics

Per-tree, per-ring, per-class metrics were computed from exact pixel overlap between the predicted binary mask and the ground truth binary mask within each ring annulus. For class c and ring k, true positives (TP), false positives (FP), and false negatives (FN) were defined as pixel counts from the intersection of predicted mask $\hat{B}_c$, ground truth mask G_c, and ring mask R_k:

$$\begin{aligned} \mathrm{TP} &= |\, \hat{B}_c \cap G_c \cap R_k \,| \\ \mathrm{FP} &= |\, \hat{B}_c \cap \overline{G_c} \cap R_k \,| \\ \mathrm{FN} &= |\, \overline{\hat{B}_c} \cap G_c \cap R_k \,| \end{aligned} \tag{3.14}$$

Five metrics were computed per class per ring [73].

Mean Absolute Error (MAE). MAE measured the absolute percentage point difference between predicted and ground truth land use coverage within a ring, providing a directly interpretable measure of coverage estimation error:

$$\mathrm{MAE} = |\, \hat{p}_c - p_c^{\mathrm{gt}} \,| \times 100 \tag{3.15}$$

where $\hat{p}_c$ and p_c^{gt} were the predicted and ground truth pixel proportions of class c within ring k, respectively. MAE served as the primary comparison metric across all experiments.

Precision. Precision measured the proportion of predicted positive pixels that were correctly classified:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{3.16}$$

Recall. Recall measured the proportion of ground truth positive pixels that were correctly predicted:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3.17}$$

F1-Score. F1-Score was the harmonic mean of Precision and Recall, providing a balanced measure of classification quality:

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3.18}$$

Intersection over Union (IoU). IoU measured the overlap between the predicted and ground truth binary masks relative to their union:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{3.19}$$

All metrics were computed from exact pixel overlap, not area approximations. Precision, Recall, F1, and IoU were set to zero when the denominator was zero, which occurred for the Building class at trees with no building pixels within the evaluation zone. All five metrics were averaged across 95 tree locations and reported as mean values per ring per class. All metrics were averaged across 95 tree locations and reported as mean values with standard deviations.

3.4.5 Statistical Analysis

To assess whether observed performance differences between pipeline configurations were statistically reliable rather than attributable to sample variation across the 95 tree locations, Wilcoxon signed-rank tests were applied to key pairwise comparisons corresponding to each research question. The Wilcoxon signed-rank test is a non-parametric paired test that evaluates whether the distribution of pairwise differences between two conditions is symmetric around zero [74]. It was selected over the paired t-test because F1-scores and MAE values are bounded and non-normally distributed, and because the presence of construction-phase outliers would violate the normality assumption required by parametric tests. The test statistic W is computed as:

$$W = \sum_{i=1}^{n} \operatorname{sgn}(x_{2i} - x_{1i}) \cdot R_i \tag{3.20}$$

where x_{1i} and x_{2i} are the paired observations for tree i, R_i is the rank of the absolute difference $|x_{2i} - x_{1i}|$, and sgn denotes the sign function. A significance level of $\alpha = 0.05$ was adopted, following the conventional threshold widely used in applied machine learning and remote sensing evaluation studies [75]. All tests were two-sided, and the null hypothesis in each case was that the median difference between paired observations was zero.

3.4.6 Evaluation Pipeline

For each tree location, the evaluation procedure was as follows:

1. CLIP classification results were loaded from `{devEUI}_2_classification.npy`.
2. SAM segment masks were loaded from `{devEUI}_2_segments.npz`.
3. Ground truth masks were loaded from `{devEUI}_2_gt.npz`.
4. A per-pixel prediction layer was constructed (Section 3.4.3).
5. Ring annulus masks were applied to isolate pixels within each buffer zone.
6. TP, FP, and FN counts were computed per class per ring from exact pixel overlap.
7. MAE, Precision, Recall, F1, and IoU were computed per class per ring.

Per-tree results were saved as `per_tree_metrics_pixelgt.csv`. Mean and standard deviation across all 95 tree locations were aggregated and saved as `final_summary_pixelgt.csv`.

4 RESULTS

4.1 Dataset and Ground Truth

The dataset comprised 95 tree locations and yielded 285 aerial image patches across three resolution variants (300, 225, and 150 DPI). Each image patch was accompanied by a corresponding `.pgw` world file. In addition, 95 pixel-level ground truth masks were available at 225 DPI, as described in Sections 3.1.4 and 3.1.5.

The distribution of land-use classes across buffer rings has already been introduced in Section 3.1.4 and is visualised in Figure 3.4. Vegetation is the dominant land-use class across all buffer rings, with its coverage decreasing from the inner to outer rings (approximately 80% → 60% → 55%). In contrast, road coverage increases with distance from the tree, indicating a stronger presence of infrastructural elements in outer buffer zones. Building coverage remains consistently low across all rings, not exceeding approximately 2% on average. Overall, the ground truth dataset exhibits a strong class imbalance, with vegetation as the dominant class and buildings forming a minor class with limited spatial presence across the study area.

4.2 Segmentation Results

To enable a consistent comparison across all segmentation configurations, the results of experiments E1–E6 are summarised in Table 4.1. In addition, the model comparison and the relationships between coverage, predicted IoU, and processing time are illustrated in Figures 4.1, 4.2, and 4.3.

4.2.1 Model Selection

ViT-H achieved the highest mean coverage at 62.5%, followed by ViT-L at 60.9%, while ViT-B performed lower at 23.7% as shown in Figure 4.1. In terms of computational cost, Vit-H required the longest processing time at 2.19 s per image, compared to 1.93 s for ViT-L and 1.66 s for ViT-B. Overall, the results show a clear increase in segmentation coverage with model size, alongside an increase in processing time. The three SAM backbone variants were evaluated on a 15-image subset at 225 DPI using default parameter settings.

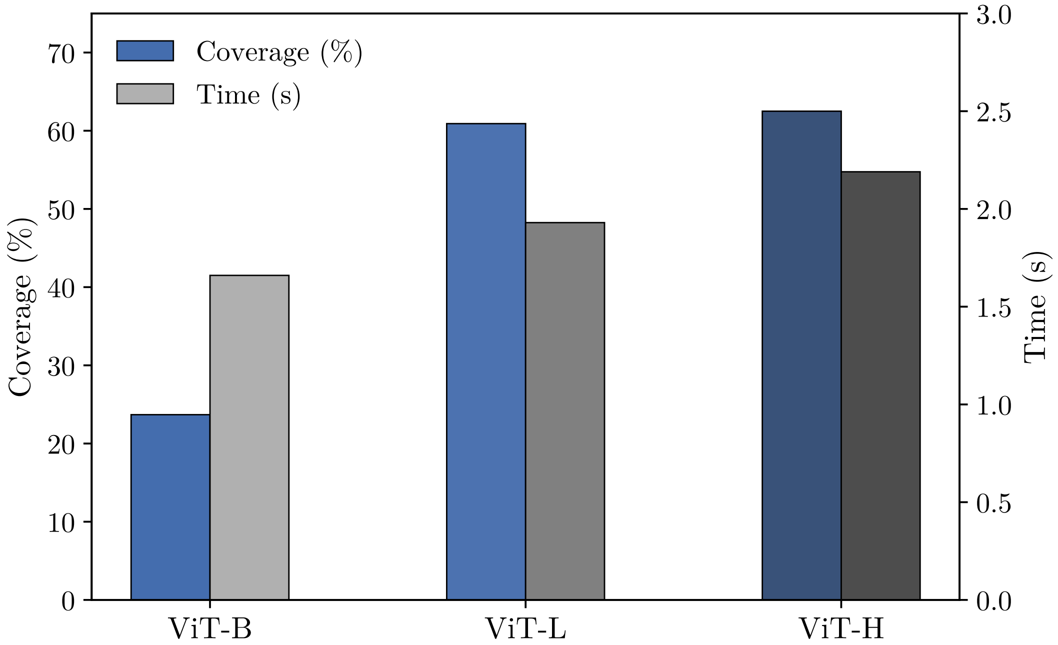

Figure 4.1: Comparison of SAM backbone variants on a 15-image subset at 225 DPI using default settings. The figure shows mean segmentation coverage in shades of blue and mean processing time per image in shades of grey.

4.2.2 Quantitative Comparison Across E1–E6

The combined comparison across all six segmentation experiments is shown in Table 4.1. The corresponding relationships between coverage and predicted IoU, and between coverage and processing time, are shown in Figure 4.2.

Exp.	**Setup**	**Seg.**	**Cov. (%)**	**Med. Area (px)**	**IoU**
E1	225 DPI, default	105.1 ± 50.1	70.5 ± 18.4	2,213	0.943
E2	225 DPI, fine-tuned	245.9 ± 96.2	93.1 ± 6.5	948	0.908
E3	300+225+150, default	61.2 ± 38.8	79.8 ± 14.3	33,082	0.938
E4	300+225+150, fine-tuned	48.1 ± 34.8	$\mathbf{96.8 \pm 2.8}$	40,469	0.907
E5	225+150, fine-tuned	50.6 ± 33.3	94.4 ± 5.2	33,960	0.907
E6	300+225, fine-tuned	47.4 ± 33.4	96.2 ± 3.3	39,932	0.912

Table 4.1: Comparison of segmentation experiments E1–E6. Values are reported as mean ± standard deviation where applicable. Median area is reported as the mean of per-image median mask areas across all 95 tree locations.

At 225 DPI, the fine-tuned configuration E2 achieved higher coverage than the default configuration E1, increasing from 70.5% to 93.1%. This change was accompanied by an increase in mean segment count from 105.1 to 245.9 per image and an increase in mean processing time from 2.2 s to 13.3 s per image. Predicted IoU was lower in E2 than in E1, changing from 0.943 to 0.908.

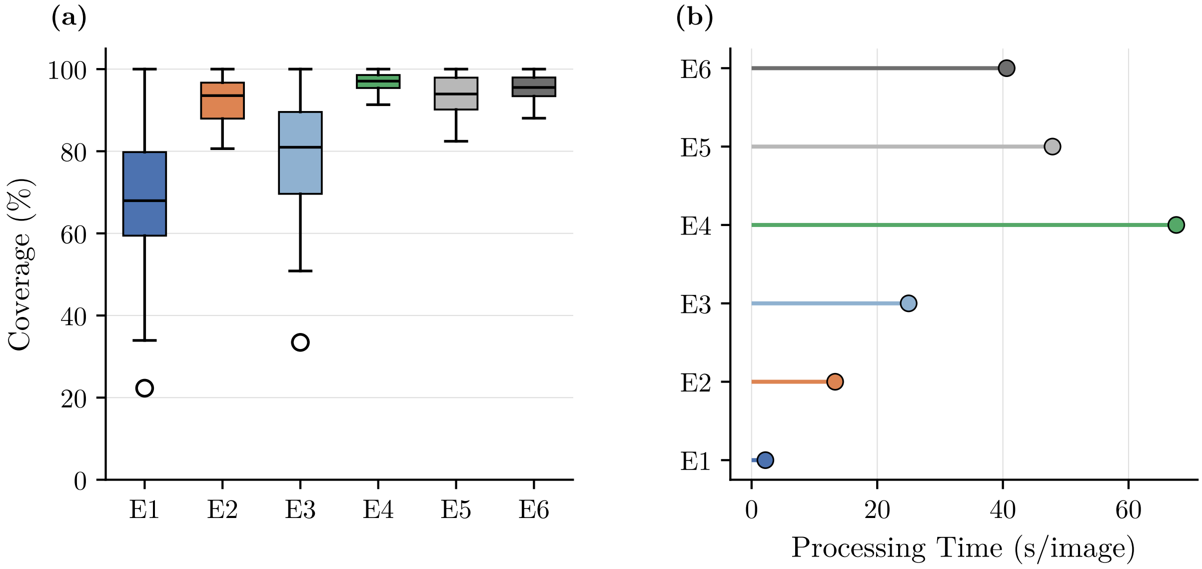

Figure 4.2: Trade-off analysis across segmentation experiments E1–E6. (a) Coverage vs predicted IoU, and (b) coverage vs processing time.

For the three-scale configurations, E3 reached a mean coverage of 79.8%, and E4 reached a mean coverage of 96.8%. E4 therefore achieved the highest mean coverage among all evaluated configurations. Relative to E3, E4 produced fewer segments and required more processing time. Predicted IoU was lower in E4 than in E3, changing from 0.938 to 0.907. Across all six configurations, E1 had the shortest mean processing time and E4 had the longest. E2 produced the highest mean segment count. E4 showed the highest mean coverage and the lowest coverage standard deviation. A Wilcoxon signed-rank test confirmed that the coverage improvement from E1 to E4 was consistent across all 95 tree locations ($W = 0$, $p < 0.001$, $n = 95$), indicating a reliable gain rather than driven by a subset of favourable cases.

4.2.3 Ablation Results

The ablation results are shown in Figure 4.3 and are also included in Table 4.1. Removing the 300 DPI scale reduced mean coverage from 96.8% in E4 to 94.4% in E5. Removing the 150 DPI scale reduced mean coverage from 96.8% in E4 to 96.2% in E6. The coverage standard deviation increased from 2.8% in E4 to 5.2% in E5 and 3.3% in E6. Mean processing time decreased from 67.6 s in E4 to 47.9 s in E5 and 40.6 s in E6.

4.2.4 Qualitative Segmentation Results

Two qualitative examples illustrate the segmentation changes across configurations. The first compares the single-scale default (E1) with the single-scale fine-tuned configuration (E2) and the second compares E2 with the multi-scale fine-tuned configuration (E4). Comparing E1 and E2 in Figure 4.4, pixel coverage increased from 41.22% to 92.66%, the

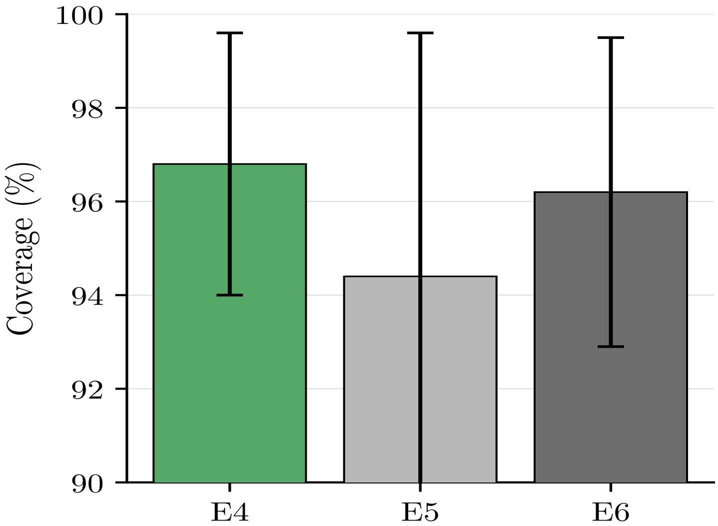

Figure 4.3: Ablation study for the multi-scale fine-tuned configuration. The figure shows mean coverage and standard deviation for E4, E5, and E6.

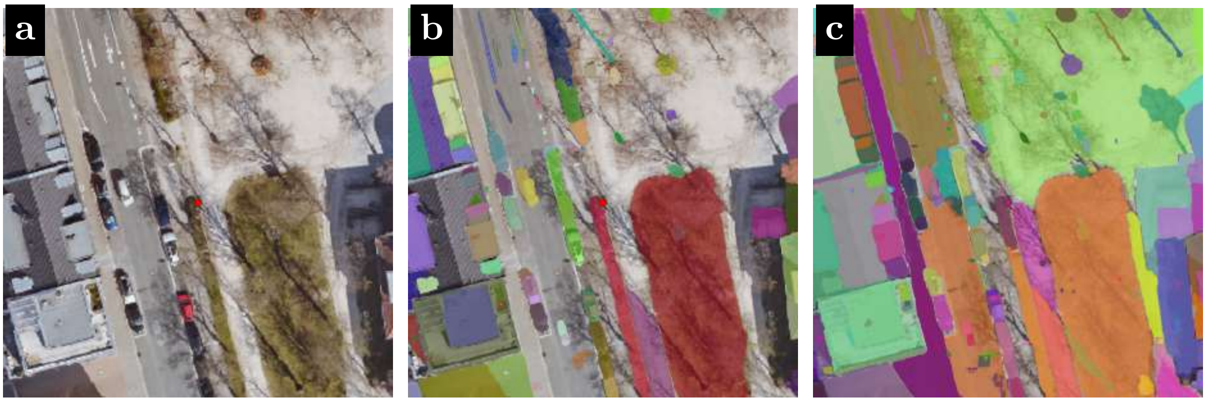

Figure 4.4: Qualitative comparison of the input image (a), and single-scale segmentation using default configuration E1 (b) and fine-tuned configuration E2 (c).

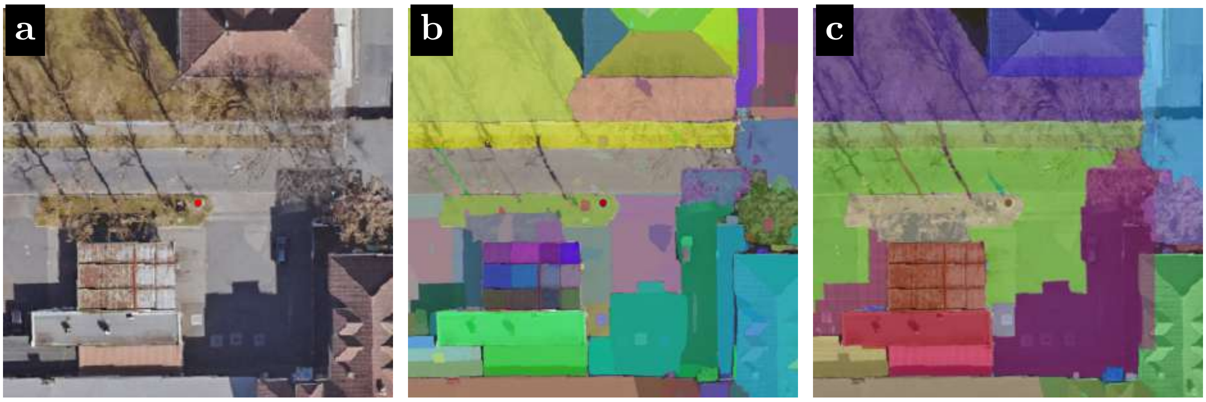

Figure 4.5: Qualitative comparison of the input image (a), fine-tuned single-scale segmentation output of E2 (b), and multi-scale fusion segmentation output of E4 (c).

segment count rise from 110 to 249, and the median segment area decreased from 1780.0 to 1327.0 pixels, resulting in higher coverage area.

For E2 and E4 in Figure 4.5, pixel coverage increased from 97.00% to 98.58%, the segment count decreased from 210 to 36, and the median segment area increased from 1097.0 to 2286.0 pixels, resulting in fewer and larger regions.

4.2.5 Selected Configuration

E4 achieved the highest mean pixel coverage across all six configurations. It also showed the lowest coverage standard deviation. E4 was selected as the segmentation input for the subsequent CLIP-based classification stage.

4.3 Classification Results

This section reports the results of the CLIP-based classification optimisation on the full Erlangen dataset. The optimisation sequence followed the five-step procedure described in Section 3.3.7. Since the study objective was to estimate land use composition within the annular buffer zones around each tree, GIS buffer evaluation was used as the basis for configuration selection. Proxy metrics were monitored during optimisation as global indicators of confidence and spatial behaviour, but they were not used as the final decision criteria. The following subsections report the results of the baseline and multi-prompt comparison, the crop-method comparison, the prompt refinement experiments, and the aggregation-mode comparison.

4.3.1 Baseline and Multi-Prompt Comparison

The baseline CLIP configuration used Prompt 0 with the zero crop-preparation strategy and single aggregation. This configuration was compared with Prompt 1 under the same crop and aggregation setting in order to assess the effect of the initial aerial multi-prompt design.

Table 4.2 summarises the GIS buffer evaluation results for both configurations. For the Vegetation class, Prompt 0 achieved higher F1-scores than Prompt 1 in all three rings. The corresponding IoU values were also higher for Prompt 0 in Rings 1, 2 and 3.

For the Road class, Prompt 1 achieved higher F1-scores and IoU values across all three rings. Road MAE was also lower for Prompt 1 in all three rings. Building performance remained low for both configurations, with F1-scores between 0.000 and 0.062 and IoU values between 0.000 and 0.051 across all rings.

Prompt 1 therefore improved Road classification at the cost of Vegetation performance relative to Prompt 0. As Prompt 1 showed sensitivity to prompt formulation, it was retained for subsequent refinement experiments rather than accepted as a final configuration.

Ring	Class	MAE (P0)	MAE (P1)	F1 (P0)	F1 (P1)	IoU (P0)	IoU (P1)
Ring 1	Vegetation	23.857	32.573	0.669	0.581	0.617	0.536
	Building	18.320	24.412	0.008	0.000	0.006	0.000
	Road	18.274	17.027	0.172	0.223	0.125	0.174
Ring 2	Vegetation	21.410	23.498	0.631	0.541	0.565	0.491
	Building	18.486	21.279	0.034	0.029	0.030	0.022
	Road	23.921	20.295	0.345	0.418	0.285	0.357
Ring 3	Vegetation	20.930	20.574	0.620	0.554	0.544	0.487
	Building	19.090	21.628	0.062	0.059	0.051	0.047
	Road	23.739	19.050	0.398	0.496	0.326	0.423

Table 4.2: GIS buffer evaluation comparison between the baseline prompt (Prompt 0) and the initial aerial multi-prompt configuration (Prompt 1), both using the zero crop-preparation strategy and single aggregation. MAE is reported in percentage points.

4.3.2 Context / Crop Method Comparison

Four crop-preparation strategies were evaluated using Prompt 1 and single aggregation on the full Erlangen dataset. Proxy metrics were computed over the full image, while GIS buffer evaluation was used for configuration selection within the tree-centred annular rings.

Strategy	Mean conf.	Low-conf. (%)	Spatial consistency	Time (s)
Zero	0.537	7.15	0.445	3.06
Highlight	0.653	2.05	0.606	2.46
Larger Crop	0.626	3.15	0.641	2.01
Dual Composite	0.716	2.02	0.772	3.92

Table 4.3: Proxy-metric comparison of the four crop-preparation strategies using Prompt 1 and single aggregation.

Table 4.3 summarises the proxy metrics for the four strategies. Dual Composite achieved the highest mean confidence (0.716) and the highest spatial consistency score (0.772), while also showing the lowest mean low-confidence percentage (2.02%). Larger Crop produced the shortest mean processing time (2.01 s per image) and the second highest spatial consistency score (0.641). In contrast, the Zero strategy showed the lowest mean confidence (0.537) and the lowest spatial consistency score (0.445). The GIS buffer evaluation showed a different ranking. Averaged across all class-ring combinations, the Zero strategy achieved the lowest mean MAE (22.52 percentage points), the highest mean F1-score (0.325), and the highest mean IoU (0.283). Highlight and Dual Composite produced similar mean MAE values of 29.12 and 29.13, respectively, while Larger Crop showed the weakest overall GIS performance, with a mean MAE of 35.77, a mean F1-score of 0.187, and a mean IoU of 0.158.

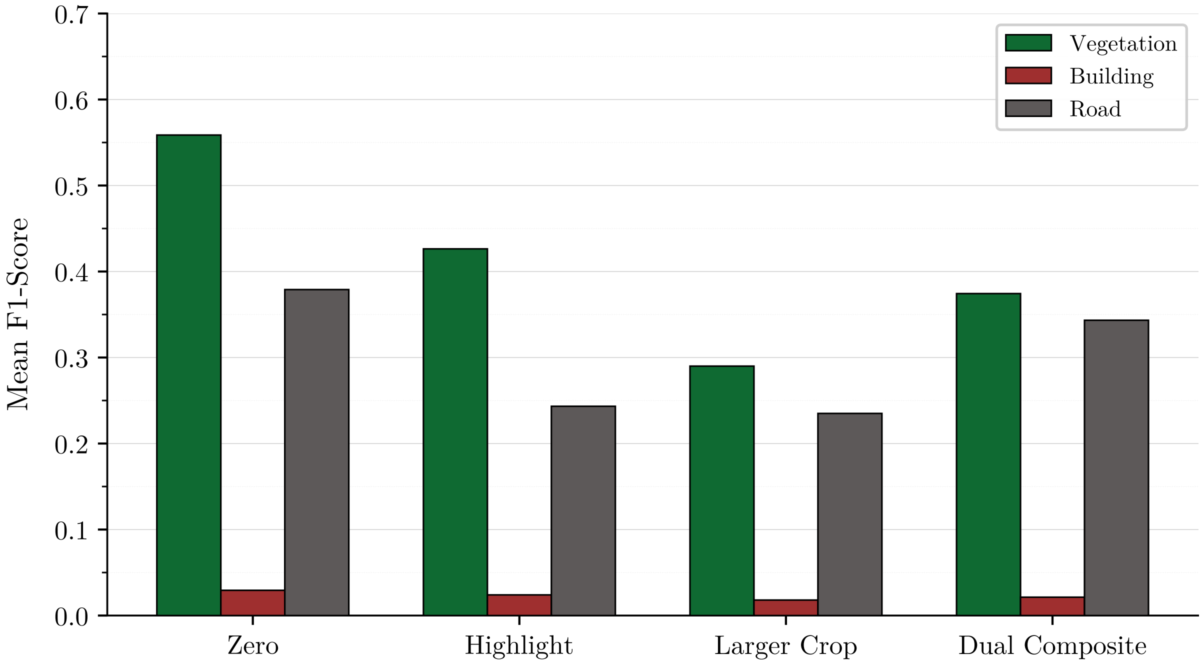

Figure 4.6: Class-wise mean F1-score across four crop-preparation strategies using Prompt 1 with single aggregation, averaged over three evaluation rings.

Figure 4.6 shows the class-level performance trends across the four crop-preparation strategies.The Zero strategy achieved the highest mean F1-score for Vegetation (0.559) and Road (0.379). Highlight performed second best for Vegetation (0.426) but poorly for Road (0.243). Dual Composite showed the second highest Road F1-score (0.343), while Larger Crop remained lowest for both classes. Building performance was low for all four strategies (0.018-0.029). Based on the GIS evaluation, the Zero strategy was selected for subsequent prompt optimisation experiments. The advantage of the Zero strategy over Dual Composite was confirmed by Wilcoxon signed rank tests across all three rings ($p < 0.001$, $n = 95$), ruling out random variation as an explanation for the lower buffer-zone error.

4.3.3 Prompt Optimisation

After the Zero crop-preparation strategy had been selected, prompt refinement was evaluated using Prompt 1.1, Prompt 1.2, and Prompt 1.3 under single aggregation.

Figure 4.7 shows the progression in class-wise F1-score across the three prompt versions. Prompt 1.3 achieved the highest Vegetation F1-score in all three rings. In Ring 1, the Vegetation F1-score increased from 0.633 for Prompt 1.1 to 0.681 for Prompt 1.2 and 0.725 for Prompt 1.3. Similar improvements were observed in Ring 2 and Ring 3, where Prompt 1.3 reached Vegetation F1-scores of 0.678 and 0.685, respectively.

For the Road class, Prompt 1.3 also produced the highest F1-scores in all three rings. The Road F1-score increased from 0.227 to 0.249 in Ring 1, from 0.409 to 0.450 in Ring 2, and from 0.474 to 0.501 in Ring 3.

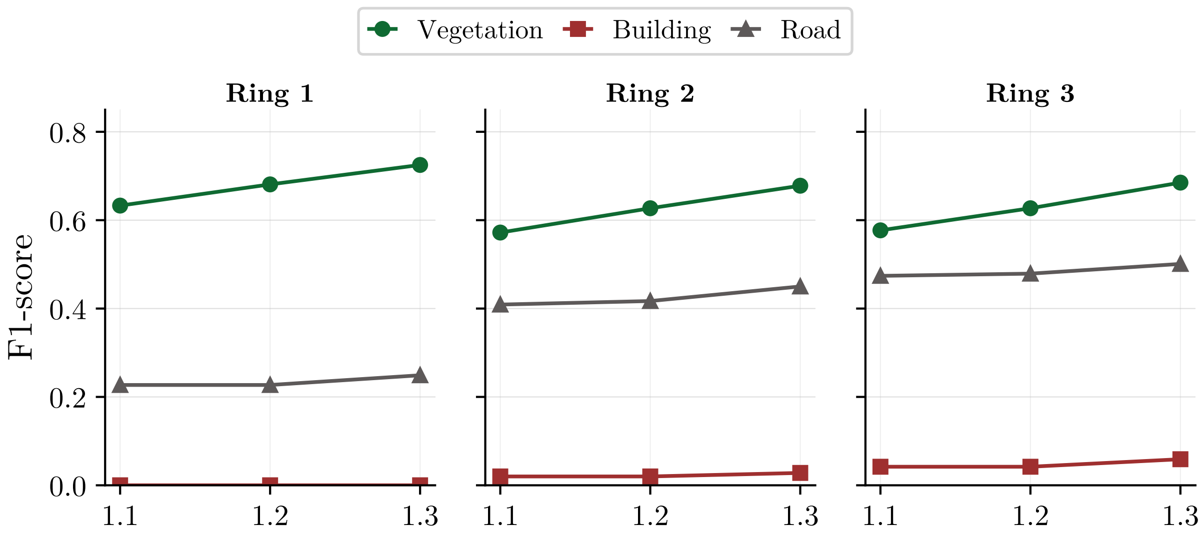

Figure 4.7: Class-wise F1-score across Prompt 1.1, Prompt 1.2, and Prompt 1.3 for the three evaluation rings using the Zero crop-preparation strategy and single aggregation.

Building performance remained low for all three prompt versions. However, Prompt 1.3 reduced Building MAE in all rings, from 17.437 to 9.982 percentage points in Ring 1, from 15.477 to 10.651 in Ring 2, and from 18.896 to 12.850 in Ring 3 relative to Prompt 1.1.

Overall, Prompt 1.3 achieved the strongest GIS buffer evaluation results among the tested prompt versions and was selected as the final prompt configuration. Wilcoxon signed-rank tests verified that Prompt 1.3 achieved higher Vegetation F1-scores than Prompt 0 across all three rings ($p < 0.002$ in all cases, $n = 95$) and higher Road F1-scores across all rings ($p < 0.002$), confirming that the improvements from aerial specific prompt design were effective across tree locations.

4.3.4 Aggregation Mode Comparison

The final optimisation step compared single and average aggregation using Prompt 1.3 with the Zero crop-preparation strategy. Table 4.4 summarises the GIS buffer evaluation results for both aggregation modes.

The two aggregation modes produced similar results across all classes and rings. For the Vegetation class, the mean F1-scores differed only slightly. In Ring 1, the mean F1-score was 0.725 for single aggregation and 0.726 for average aggregation. In Ring 2, the corresponding values were 0.678 and 0.670, and in Ring 3 they were 0.685 and 0.680. Vegetation MAE values were also similar, with single aggregation showing slightly lower MAE in Rings 2 and 3.

For the Road class, single aggregation achieved slightly higher F1-scores in all three rings. For the Building class, both aggregation modes remained weak, and the absolute differences between the two aggregation modes were small across all reported metrics.

Ring	Class	MAE (single)	MAE (avg.)	F1 (single)	F1 (avg.)
Ring 1	Vegetation	20.647	20.702	0.725	0.726
	Building	9.982	9.761	0.000	0.000
	Road	13.678	13.623	0.249	0.248
Ring 2	Vegetation	18.016	19.037	0.678	0.670
	Building	10.651	9.939	0.028	0.028
	Road	17.463	17.933	0.450	0.448
Ring 3	Vegetation	16.253	17.140	0.685	0.680
	Building	12.850	12.432	0.059	0.065
	Road	18.056	18.479	0.501	0.495

Table 4.4: GIS buffer evaluation comparison between single and average aggregation using Prompt 1.3 and the Zero crop-preparation strategy. MAE is reported in percentage points.

Overall, single aggregation showed slightly stronger results for Road and similar results for Vegetation, while average aggregation yielded only minor differences. Single aggregation was therefore selected as the final aggregation mode.

4.3.5 Selected Configuration and Final GIS Summary

Based on the sequential optimisation results, the final CLIP classification configuration used Prompt 1.3, the Zero crop-preparation strategy, and single aggregation. Prompt 1.3 achieved the strongest GIS buffer evaluation results among the tested prompt versions, the Zero strategy achieved the best overall agreement with ground truth among the four crop-preparation methods, and single aggregation produced slightly stronger results than average aggregation while remaining similar for Vegetation. Figure 4.8 summarises

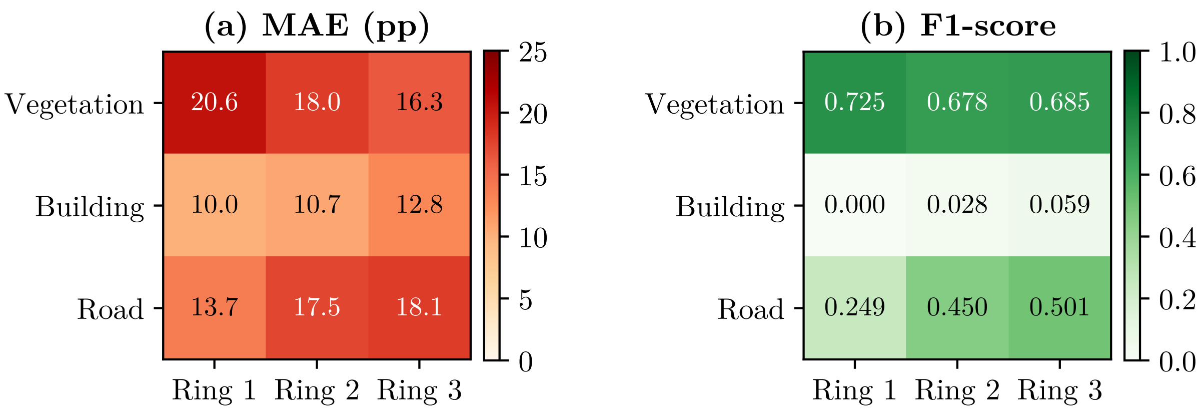

Figure 4.8: GIS buffer evaluation of the selected CLIP configuration across the three annular rings. Panel (a) shows MAE and panel (b) shows F1-score for Vegetation, Building, and Road.

the final GIS buffer evaluation results of the selected configuration across the three annular rings. Panel (a) shows that Vegetation MAE decreased from Ring 1 to Ring 3, whereas Road MAE increased across the three rings. Building MAE remained relatively stable. Panel (b) shows that Vegetation achieved the strongest F1-scores across all rings, Road improved from Ring 1 to Ring 3, and Building remained low throughout the evaluation.

For Vegetation, the F1-scores were 0.725, 0.678, and 0.685 in Rings 1, 2, and 3, respectively. The corresponding MAE values were 20.647, 18.016, and 16.253 percentage points. For Road, the F1-scores increased from 0.249 in Ring 1 to 0.450 in Ring 2 and 0.501 in Ring 3. Building remained the weakest class, with F1-scores of 0.000, 0.028, and 0.059 across the three rings.

4.4 Cross-Dataset Generalisation

The selected pipeline configuration (E4 segmentation, Prompt 1.3, Zero crop, Single aggregation) was evaluated on Aerial 2025 and Google Satellite imagery as described in Section 3.1.2. This section reports quantitative performance comparison across the three image sources, assessing the pipeline's robustness to temporal, seasonal, and sensor variations without any retraining or parameter adjustment.

4.4.1 Quantitative Performance Comparison

Table 4.5 presents the MAE comparison across all three image sources. Google Satellite achieved the lowest MAE across all three rings for Vegetation (13.3, 14.4, 13.6) and Building (5.5, 5.2, 6.0) percentage points. For Road, it also achieved the lowest MAE in all three rings, with values of 10.9, 17.3, and 17.2 percentage points.

Ring	**Class**	**Aerial 2020**	**Aerial 2025**	**Google Satellite**
		MAE (percentage points, mean ± std)		
Ring 1	Vegetation	20.6 ± 29.0	27.7 ± 32.1	$\mathbf{13.3 \pm 19.5}$
	Building	10.0 ± 22.5	7.6 ± 15.6	$\mathbf{5.5 \pm 15.2}$
	Road	13.7 ± 22.4	23.8 ± 30.1	$\mathbf{10.9 \pm 17.7}$
Ring 2	Vegetation	18.0 ± 26.3	16.9 ± 22.2	$\mathbf{14.4 \pm 20.8}$
	Building	10.7 ± 22.3	6.0 ± 14.4	$\mathbf{5.2 \pm 15.5}$
	Road	17.5 ± 23.9	20.1 ± 23.5	$\mathbf{17.3 \pm 22.9}$
Ring 3	Vegetation	16.3 ± 21.1	14.2 ± 17.0	$\mathbf{13.6 \pm 16.7}$
	Building	12.9 ± 22.8	6.9 ± 13.9	$\mathbf{6.0 \pm 12.3}$
	Road	18.1 ± 20.3	17.3 ± 18.9	$\mathbf{17.2 \pm 20.6}$

Table 4.5: MAE comparison across different image sources. Bold values indicate the lowest MAE for each class ring combination, with standard deviations reported across 95 tree locations. The pipeline was optimised on Aerial 2020 and applied unchanged to Aerial 2025 and Google Satellite imagery.

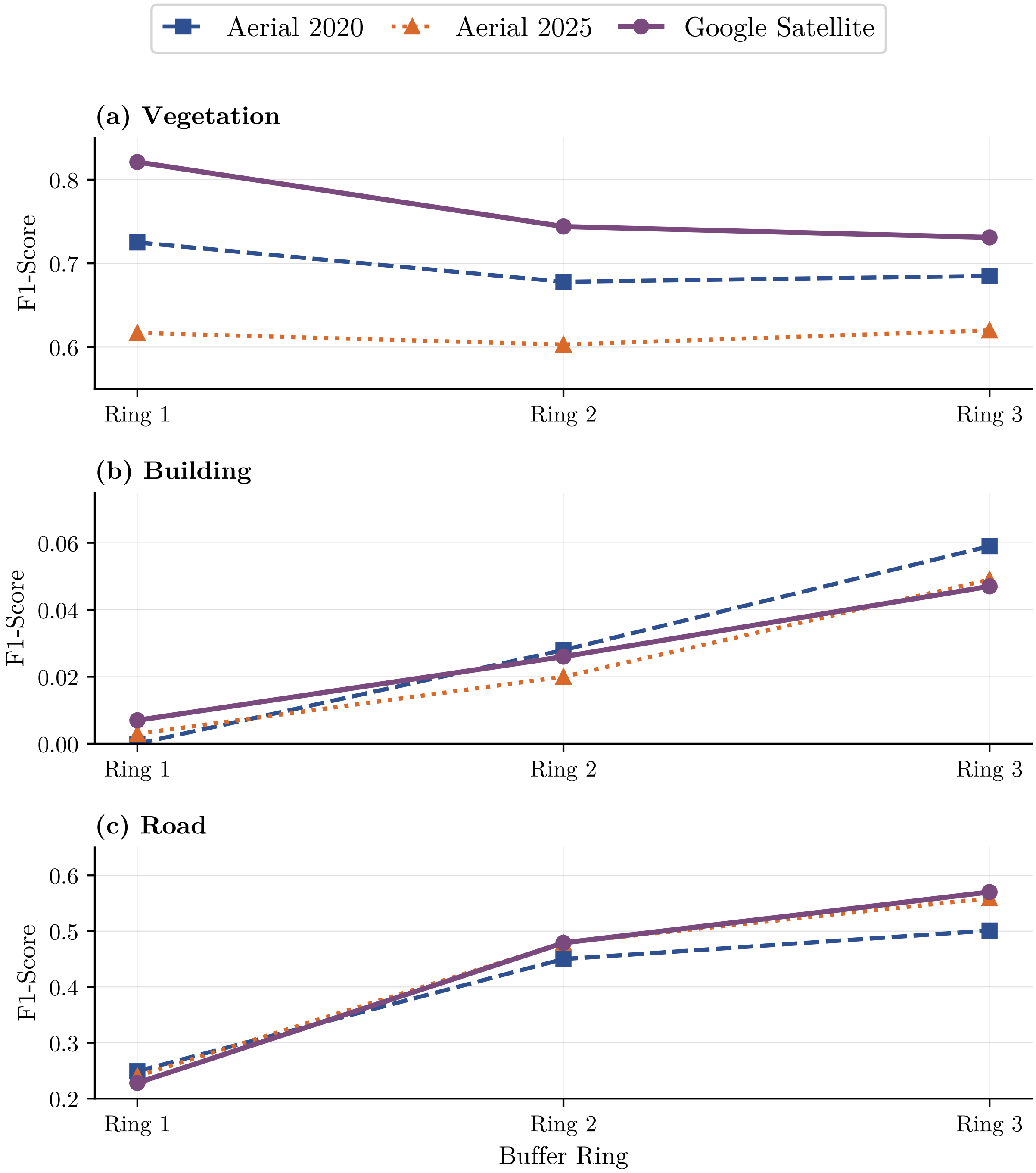

Figure 4.9: F1-score comparison across image sources showing performance trends from inner to outer buffer rings. Panel (a) shows Google Satellite achieved consistently higher vegetation F1-scores across all rings. Panel (b) shows building classification remained near zero for all datasets. Panel (c) shows road F1-scores improved with distance from the tree centre, with all three datasets converging in Ring 3.

Aerial 2025 showed degraded vegetation performance in Ring 1 compared to Aerial 2020 (MAE increased from 20.6 to 27.7 percentage points) but improved performance in Rings 2 and 3 (MAE decreased from 18.0 to 16.9 and from 16.3 to 14.2 percentage points, respectively). Building MAE improved across all rings in Aerial 2025 compared to Aerial 2020. Road MAE increased in Ring 1 (from 13.7 to 23.8 percentage points) but remained relatively stable in outer rings. Wilcoxon signed-rank tests showed a confirmed difference between Aerial 2020 and Google Satellite in Ring 1 only ($W = 1412.5$, $p = 0.049$), while outer rings did not reach the significance threshold ($p = 0.112$ and $p = 0.426$ respectively).

Figure 4.9 illustrates the F1-score trends across buffer rings for all three datasets. Google Satellite achieved the highest vegetation F1-scores across all rings (0.821, 0.744, 0.731), outperforming both Aerial 2020 (0.725, 0.678, 0.685) and Aerial 2025 (0.617, 0.603, 0.620). Road F1-scores showed an upward trend with increasing distance from the tree centre across all datasets, with Aerial 2025 reaching 0.559 in Ring 3 compared to 0.501 for Aerial 2020 and 0.570 for Google Satellite. Building F1-scores remained below 0.059 across all datasets and rings.

4.4.2 Qualitative Comparison

For all figures in this subsection, panels are arranged as follows: original image (left), SAM segmentation (middle left), classification overlay (middle right), and performance bar chart (right). Classification colours indicate: green = Vegetation, grey = road, red = Building. Bar charts show ground truth (light shade bars) and predictions (solid bars) per buffer ring. Buffer rings at 2.5 m, 5.0 m, and 7.5 m are shown as white circles.

Figure 4.10 presents a vegetation-dominated site where classification remained consistent across all three image sources. All three datasets classified the buffer zones as vegetation (>95% in all rings). The building structure visible in the lower portion of each image was correctly identified across datasets. SAM segmentation produced distinct object boundaries in all three cases, as indicated by the clearly separated colour-coded regions in the segmentation panels. The transition from sparse canopy in Aerial 2020 to full foliage in Aerial 2025 and Google Satellite did not affect classification results.

Figure 4.11 presents a site where a building was under construction in 2020 and completed by 2025, with vegetated spaces clearly visible. Aerial 2020 (top row) captured the site during active construction, showing bare ground and incomplete structures. SAM segmented the entire image into only 3 to 4 segments, with the largest segment covering the construction area along with surrounding vegetation and partially built structures. When this segment was passed to CLIP, it was misclassified as Building (red), showing Building proportions of approximately 100% across all rings (red bars in bar chart).

By 2025 (middle row), the site showed completed development with established vegetation and paved infrastructure. SAM segmentation produced 21 distinct segments, reflecting the site's structural complexity. Accordingly, classification showed Vegetation in all rings (approximately 90%, 70%, and 65% in Rings 1, 2, and 3 respectively), with Road classification in the western portion. Similarly, Google Satellite (last row) showed

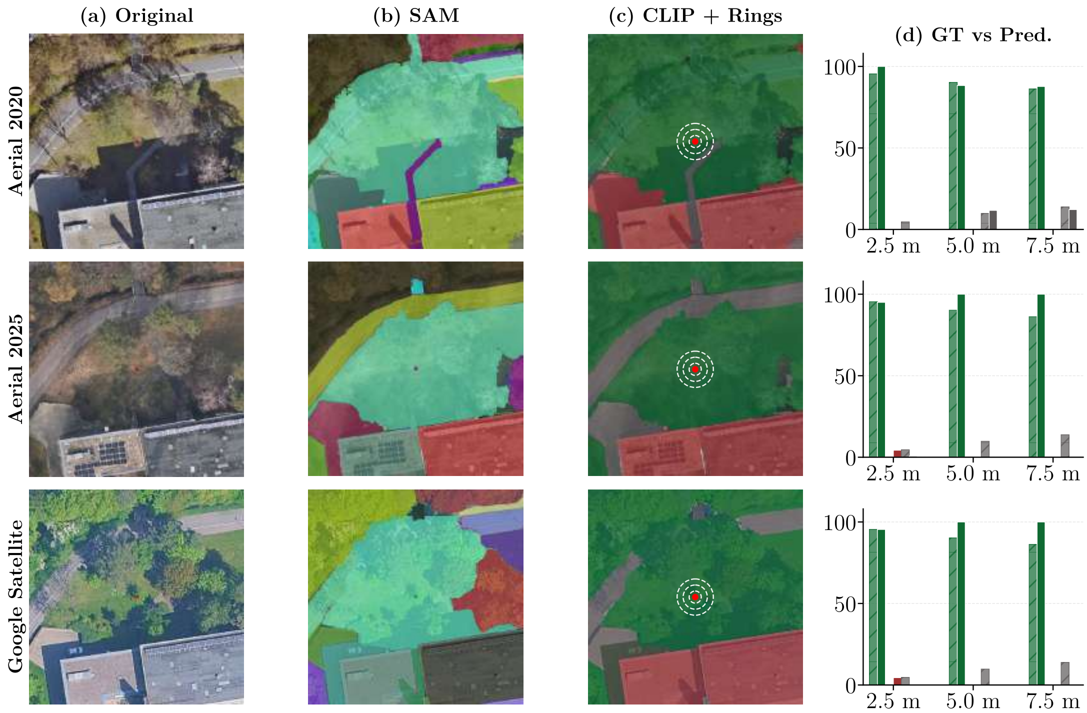

Figure 4.10: Consistent classification across all three image sources for a stable vegetation scene. Vegetation classification exceeded 95% across all rings and datasets.

classification patterns to Aerial 2025, with Vegetation class of approximately 95%, 75%, and 70% across the three rings.

Figure 4.12 shows a failure case where building shadows caused misclassification across all three image sources. Diagonal shadows were segmented as single masks rather than being separated from the underlying land cover. These shadow masks were subsequently misclassified as Road or Building, depending on the image source and lighting conditions.

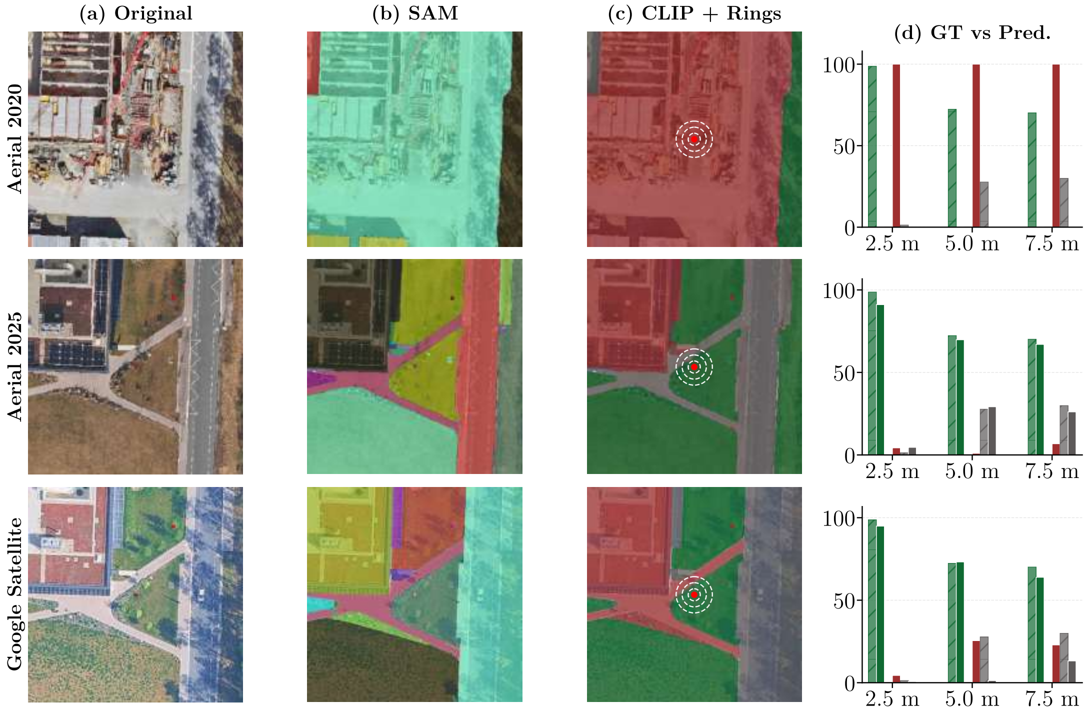

Figure 4.11: Temporal land-cover change from construction to completed development. Aerial 2020 misclassified the site as Building (100% red bars), while 2025 and Google Satellite correctly identified established vegetation (70–95% green bars).

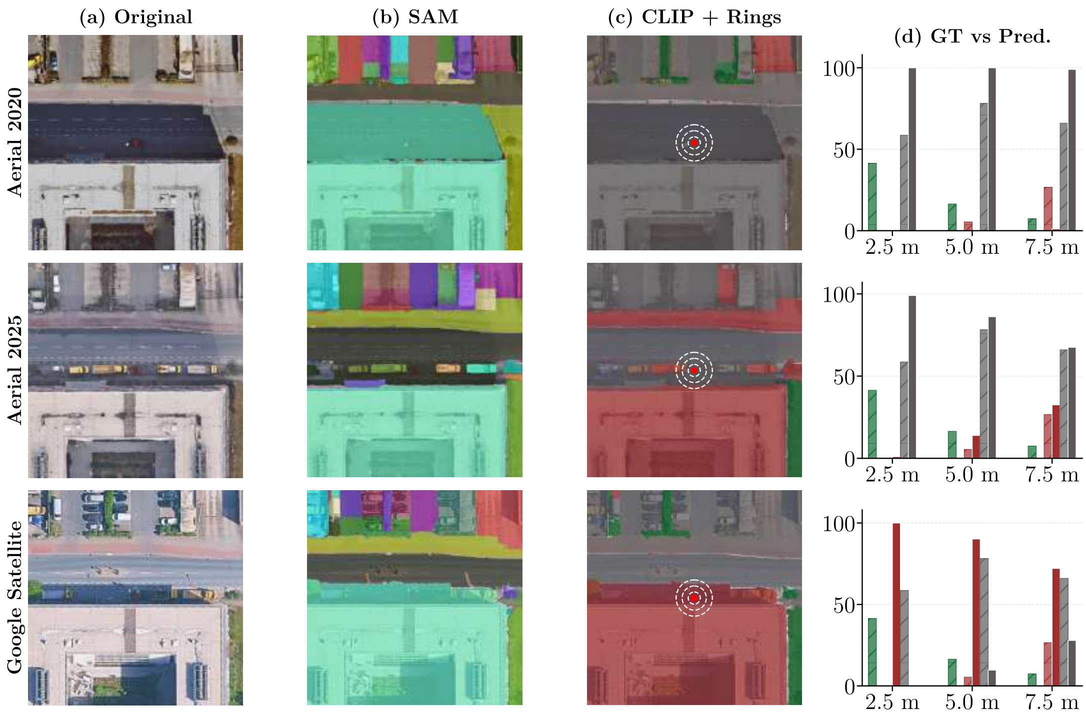

Figure 4.12: Failure case demonstrating how building shadows are segmented as single masks and misclassified across all three image sources.

5 DISCUSSION

This chapter discusses the main findings of the proposed SAM $\rightarrow$ CLIP $\rightarrow$ GIS pipeline for tree-centred urban land composition estimation from aerial imagery. The results showed that multi-scale fine-tuned SAM segmentation produced more complete and structurally coherent regions than the baseline configurations, providing more suitable inputs for downstream classification. For CLIP-based zero-shot classification, the selected configuration combined Prompt 1.3, the Zero crop-preparation strategy, and Single aggregation, yielding strong performance for Vegetation, moderate performance for Road, and consistently weak performance for Building. Cross-dataset evaluation further showed that the pipeline generalised best to stable vegetation-dominated scenes, particularly on Google Satellite imagery, while construction phases and strong shadow conditions remained systematic failure cases.

5.1 Motivation for the Proposed Approach

Prior experiments on the same Erlangen dataset evaluated SegFormer and SAM+Heuristics for aerial image segmentation using whole-image pixel-level metrics. SAM+Heuristics achieved 59.6% pixel accuracy across all image regions, demonstrating zero-shot segmentation capability but relying on manually defined heuristic rules for classification. Three key limitations motivated the present work. First, heuristic classification provides no semantic understanding of scene content. Second, whole-image evaluation lacks the spatial context required for tree-centric analysis. Third, pixel accuracy does not translate into actionable land composition metrics suitable for urban planning applications [29].

The current work addresses these gaps by replacing heuristic classification with CLIP-based zero-shot semantic classification, introducing GIS buffer analysis to incorporate spatial context, and deriving tree-level land composition metrics. SAM is retained as the segmentation backbone due to its strong zero-shot segmentation capability; however, the classification and evaluation pipeline has been systematically redesigned. Due to differences in evaluation scope and the absence of consistent ground truth across settings, direct comparison with prior results is not appropriate. Instead, SAM and CLIP are evaluated against their respective baseline configurations within the proposed framework, where default parameter settings serve as reference points for assessing the impact of the introduced modifications.

5.2 Segmentation Configuration Selection

ViT-H was selected as the SAM backbone based on its increased pixel coverage compared to ViT-L and ViT-B, as shown in figure 4.1, despite its higher runtime, since segmentation completeness directly determines the quality of downstream classification input. The backbone selection was based on a 15-image subset, which represents a limited sample of the full dataset. However, the subsequent application of ViT-H across all 95 tree locations in experiments E1–E6 confirmed that the coverage advantage observed in the subset held consistently at the full dataset scale, supporting the validity of the selection.

Fine-tuning consistently improved pixel coverage across both single-scale and multi-scale configurations (Table 4.1). At 225 DPI, coverage increased from 70.5% (E1) to 93.1% (E2), accompanied by a rise in mean segment count from 105.1 to 245.9 per image. This improvement is visually confirmed in Figure 4.4, where E2 covers a larger portion of the scene compared to E1. Since unsegmented regions provide no input for the subsequent CLIP-based classification stage, coverage was prioritised over predicted IoU (Figure 4.2).Although predicted IoU decreases slightly with fine-tuning (0.943 $\rightarrow$ 0.908 for E1 $\rightarrow$ E2; 0.938 $\rightarrow$ 0.907 for E3 $\rightarrow$ E4), it reflects SAM's internal confidence rather than an external evaluation metric and should not be over-interpreted.

However, high coverage alone is insufficient for reliable classification. The sharp rise in segment count to 245.9 per image in E2, combined with a median mask area of only 948 pixels (Table 4.1), indicates a highly fragmented output across the dataset. In contrast, E4 reduced the mean segment count to 48.1 per image while increasing the median mask area to 40,469 pixels, confirming that multi-scale merging produced fewer and geometrically larger regions rather than simply suppressing small masks. This fragmentation behaviour is visually confirmed in Figure 4.5, where a single building was divided into four separate masks and vegetation regions were similarly fragmented in E2 whereas E4 represents the same scene with clearly fewer and more coherent regions.

The multi-scale approach addressed this fragmentation problem through two complementary mechanisms. First, the three DPI levels provide information at different spatial scales: the 300 DPI images preserve fine boundaries and local detail, while the 150 DPI images provide broader contextual information that stabilises segmentation over larger regions. Second, the merging strategy consolidates overlapping and adjacent masks across resolutions into fewer, larger, and more coherent regions. As shown in Table 4.1, E4 achieved comparable coverage to E2 (96.8% versus 93.1%) while reducing the mean segment count from 245.9 to 48.1 per image. This interpretation is visually confirmed in figure 4.5, where E4 represents the same scene with clearly fewer and larger regions compared to E2, without a meaningful loss in coverage. The IoU fusion threshold of 0.60 was selected from three candidate values (0.30, 0.60, 0.99) rather than through a systematic grid search. The three values were chosen to represent qualitatively distinct merging behaviours: aggressive merging at 0.30 caused spatially adjacent but distinct objects to be incorrectly merged, while conservative merging at 0.99 missed duplicate detections of the same object across resolution levels. The intermediate value of 0.60 avoided both failure modes, suggesting it represented a stable operating point rather than a sharp optimum requiring finer calibration.

The ablation results confirm the individual contribution of each scale (figure 4.3, table 4.1). Removing the 300 DPI scale (E5) caused a larger drop in coverage (94.4%) and a greater increase in standard deviation (5.2%) than removing the 150 DPI scale (E6), which yielded 96.2% coverage with a standard deviation of 3.3%. This suggests that the finer spatial representation of 300 DPI contributed more strongly to recovering missed regions, while the 150 DPI scale provided useful but less critical complementary context. Although E6 required less processing time (40.6 s versus 67.6 s for E4), the lower coverage standard deviation of E4 indicates more consistent segmentation across varying scene compositions. E4 was therefore selected over E6 despite the higher computational cost.

Segmentation quality was assessed primarily via coverage-based and SAM-internal metrics rather than full pixel-level accuracy against ground truth for every experimental configuration. The interpretation of segmentation quality is therefore focused on completeness, consistency, and structural behaviour rather than on detailed boundary accuracy. In summary, the results show that multi-scale segmentation fusion improved both coverage and segment quality, with coverage increased from 70.5% (E1, single-scale default) to 96.8% (E4, multi-scale fine-tuned) while fragmentation decreased, producing fewer and larger segments that provide more reliable inputs for zero-shot classification. The Wilcoxon test result ($p < 0.001$) supports this conclusion, indicating that the coverage gain was consistent across all 95 tree locations instead of being supported by only a few positive cases. This is expected because single-scale and multi-scale segmentation work differently. By combining information from three resolution levels, the pipeline captures object boundaries that are invisible at any single scale. This improves a result of the method itself rather than a dataset-specific effect.

5.3 CLIP-Based Classification

The selected CLIP configuration (Prompt 1.3, zero crop, single aggregation) shows different performance across classes. It works well for vegetation, performs moderately for roads, and performs poorly for buildings. An important finding is that proxy metrics computed over the whole image do not always match the GIS-based metrics computed within buffer rings. In some cases, configurations that give higher confidence at the image level actually perform worse in the buffer-zone evaluation. This difference was important in how the final optimisation results were interpreted.

Prompt design affected classification performance and directly addressed the second research question. Prompt 0 used generic terms (“a photo of a road”) that activated CLIP’s ground-level associations, while Prompt 1 incorporated aerial-specific descriptors (“asphalt road from above”). Progressive refinement through Prompts 1.1–1.3 added colour diversity for vegetation (“green or brown”) and functional emphasis for roads (“pavement for vehicles”), with Prompt 1.3 achieving the strongest results. However, these gains demonstrate diminishing returns: initial viewpoint adaptation provided the largest benefit, while subsequent refinement yielded progressively smaller improvements, consistent with findings in the broader vision-language literature [37, 38]. In the reported comparisons, aerial-specific prompt design improved Road classification by up to 10.5 percentage points and Vegetation by up to roughly 9 percentage points from generic to refined descriptors, while Building remained near zero (F1 < 0.06) regardless of prompt formulation. This indicates that text descriptions alone cannot overcome fundamental visual ambiguities when CLIP's learned concepts mismatch the overhead viewpoint [31, 15]. Aerial-specific prompt adaptation therefore improved classification for visually distinctive and well-represented classes but showed clear limits for ambiguous ones. The per-tree analysis ($p < 0.002$ across all rings for both Vegetation and Road classes) shows that these improvements were consistent across tree locations and were not limited to only a few well-classified examples.

The crop-preparation experiments revealed that context-enhanced methods (Highlight, Larger Crop, Dual Composite) achieved higher whole-image confidence (0.716 for Dual Composite versus 0.537 for Zero) yet Zero cropping achieved the lowest buffer-zone error (MAE 22.52 versus 29.13 percentage points). This divergence suggests that higher confidence on the full image did not reflect better accuracy within the small regions of interest. CLIP was designed for image-level classification rather than spatial reasoning about small segments [31], meaning context-enhanced inputs likely introduced visual patterns from unannotated areas that increased global confidence while biasing predictions inside the buffer zones. Additionally, the ground truth covered only 3% of each image area, so surrounding context could reflect systematic environmental differences rather than actual buffer-zone land cover. Composite and highlighted inputs also likely fell outside CLIP's natural-image training distribution [45, 42]. The modest difference between single and average aggregation (F1 differences < 0.01) confirmed that this step was less critical than prompt design and crop preparation.

The classification results broadly followed the ground truth composition of the buffer zones, which directly addressed the third research question. Ground truth shows vegetation decreasing from 83.3% in Ring 1 to 54.6% in Ring 3, while road presence increases from 16.6% to 43.4%. Vegetation performed strongly (F1 0.68–0.73, MAE 16–21 percentage points) due to its dominant representation and distinctive spectral and textural signatures at 20 cm resolution, while Road classification improved with distance from the tree centre as larger contiguous segments in outer rings provided more reliable classification targets. Buffer radius therefore affected the measured land composition and the classification accuracy, a pattern that held consistently across rings. However, roads exhibited high intra-class variability (highways, residential streets, parking lots) and frequently adjoined buildings, creating mixed segments when SAM boundaries did not align precisely with road edges.

Building classification failed within the buffer zones despite successful detection visible outside the evaluated rings, indicating that specific conditions within the small buffer zones prevented accurate classification rather than a general inability to recognise buildings. Building rooftops lack distinctive overhead features, appearing as flat surfaces that resemble roads and parking lots, and CLIP's pre-training on ground-level photography emphasises vertical facades rather than rooftop appearance [31, 10]. SAM boundary errors in the confined buffer zones further compounded this by creating mixed segments containing both rooftop and asphalt materials. Importantly, this failure differs from class imbalance effects in supervised learning. Since this thesis employed zero-shot classification without any training on the Erlangen dataset, CLIP did not learn from the local class distribution, so the rarity of buildings in the buffer zones (mean coverage 1–4%) should not directly affect classification behaviour. Cases where buildings constituted up to 33.6% of Ring 2 yet were still misclassified confirm that the failure stems from visual ambiguity and SAM boundary errors rather than insufficient representation. Addressing this would likely require fine-tuning on aerial imagery or incorporating auxiliary spatial data such as cadastral building footprints.

5.4 Cross-Dataset Generalisation

The cross-dataset evaluation applied the pipeline, trained and optimised on Aerial 2020, to Aerial 2025 and Google Satellite imagery without any parameter changes, directly addressing the fourth research question. The three datasets differ in acquisition year, seasonal conditions, and sensor processing, making the evaluation inherently challenging due to confounded factors that cannot be fully isolated.

Overall, Google Satellite achieved the best performance across all datasets, with lower vegetation MAE (13.3–13.6 percentage points) and higher F1-scores (0.73–0.82) compared to the aerial datasets (0.60–0.73). This is likely due to more consistent radiometric processing and stronger seasonal vegetation signals. Aerial 2025 showed intermediate performance, while Aerial 2020 performed least reliably, particularly in early growth or construction-related scenes.

The outer buffer rings did not show statistically significant differences between datasets ($p > 0.1$). This can be explained by a combination of factors. First, construction-phase areas in Aerial 2020 introduced localised variability in Ring 1, where bare soil and incomplete structures dominated the immediate surroundings of several trees. Second, some scenes contain very small vegetation patches embedded within dominant land-cover types such as Roads and Buildings. These patches are often merged into larger segments by SAM, leading to classification ambiguity. Finally, outer buffer regions are generally more homogeneous across all datasets, reducing the potential for measurable differences in land-cover composition.

The qualitative analysis further highlights three representative behaviours. Vegetation-dominated scenes showed stable classification across all datasets, demonstrating strong robustness when clear land-cover structure is present. In contrast, a construction-phase site in Aerial 2020 was misclassified as Building due to incomplete structures and exposed ground, while the same location in later datasets was correctly classified as Vegetation after development completion. This indicates that the pipeline is sensitive to actual surface conditions, making it suitable for monitoring urban change but less reliable during transitional construction phases. A third case showed that strong building shadows can lead to segmentation fragmentation and incorrect classification, as shadowed regions are often treated as separate objects by SAM and then misinterpreted by CLIP.

Overall, the results show that the pipeline generalises well to stable vegetation-dominated environments, with Google Satellite performing most consistently. However, construction phases and strong shadow conditions remain systematic failure cases that cannot be fully resolved through parameter tuning alone and may require additional spatial priors or preprocessing.

5.5 Limitations

Several limitations should be considered when interpreting the results of this thesis.

The segmentation evaluation relied primarily on coverage-based and SAM-internal quality metrics rather than full pixel-level segmentation accuracy against exhaustive ground truth for every experimental configuration. The interpretation of segmentation quality is therefore focused on completeness, consistency, and structural behaviour rather than on detailed class-specific boundary accuracy. While the reduced segment count in E4 suggests improved suitability for CLIP-based classification, this relationship was not evaluated independently as a separate optimisation objective. The direct influence of segmentation structure on downstream classification accuracy therefore remains unquantified.

The ground truth annotations were designed to support compositional estimation within buffer zones, providing pixel-level land-cover labels within the three annular rings around each tree. This matches the application objective of estimating land composition around urban trees and is appropriate for the GIS buffer evaluation used throughout this thesis. However, it means that pixel-level classification accuracy across the full image could not be assessed. Evaluating how precisely CLIP classifies individual segments or how accurately SAM delineates object boundaries across the entire scene would require exhaustive pixel-level annotations beyond the buffer zones. The evaluation is therefore specific to the tree-centred compositional context and cannot be directly interpreted as general urban land-cover mapping accuracy. The qualitative examples selected to illustrate pipeline behaviour represent individual cases and cannot capture the full variability across all 95 tree locations. They should therefore be interpreted as supporting visual evidence rather than as a substitute for the quantitative comparisons.

The pipeline performed more reliably for dominant and visually distinctive classes than for rare or visually ambiguous ones. Building classification failed consistently within the buffer zones, likely due to visual similarity with road surfaces and domain mismatch in CLIP's pre-training. A related observation from the qualitative results is that small vegetation patches such as narrow grass strips between pathways or road edges were frequently either not segmented by SAM or misclassified by CLIP. These regions often produce image patches that are too small for SAM to detect as distinct segments at the evaluated resolutions. When they are included in larger mixed segments, their vegetation signal is diluted by surrounding road or building materials. This means the pipeline may systematically underestimate vegetation in fragmented or fine-grained urban scenes where green space occurs in narrow strips rather than contiguous areas.

The prompt design optimised in this thesis was developed specifically for the Erlangen dataset and the Aerial 2020 imagery. Each descriptor was selected based on domain knowledge of the local urban landscape, and the set of terms that performed well here may not transfer directly to cities with different vegetation types, building materials, road surfaces, or climatic conditions. Similarly, the SAM hyperparameter configuration was tuned on the same dataset and may require adjustment for imagery with different resolution, radiometry, or scene composition. Both the prompt design and hyperparameter tuning therefore represent dataset-specific optimisation steps rather than universally applicable configurations.

The cross-dataset evaluation is constrained by the confounded nature of the three image sources. Aerial 2020 versus Aerial 2025 combines temporal land-cover change with seasonal variation, while Google Satellite introduces different sensor processing alongside seasonal state. Performance differences between datasets therefore cannot be attributed to any single factor, and the relative contribution of temporal change, seasonal state, and sensor characteristics remains unclear. This limits the conclusions that can be drawn about which specific aspect of image acquisition most strongly affects pipeline performance.

Wilcoxon signed-rank tests were performed for the key configuration comparisons corresponding to each research question. The results confirmed reliable differences for RQ1 (E1 vs E4 coverage, $p < 0.001$), RQ2 (Prompt 0 vs Prompt 1.3 Vegetation and Road F1, $p < 0.002$ across all rings), and RQ3 (Zero vs Dual Composite Vegetation MAE, $p < 0.001$). For RQ4, the difference between Aerial 2020 and Google Satellite reached the significance threshold only in Ring 1 ($p = 0.049$) and not in outer rings ($p > 0.1$), indicating that the cross-dataset performance gap should be interpreted with caution and is partly attributable to construction-phase outliers in the Aerial 2020 dataset rather than a consistent systematic difference across all locations.

6 CONCLUSION

This thesis developed and evaluated a zero-shot pipeline combining SAM-based segmentation, CLIP-based classification, and GIS buffer analysis for estimating land composition around urban trees in Erlangen, Germany. The results showed that the proposed approach can produce useful tree-centred land composition estimates without any domain-specific training on the target dataset. Multi-scale segmentation fusion improved both coverage and segment coherence, with the selected three-scale fine-tuned configuration (E4) achieving 96.8% pixel coverage while reducing fragmentation substantially compared with the single-scale fine-tuned baseline. For classification, aerial-specific prompt design improved performance for visually distinctive classes, with strong results for Vegetation and moderate results for Road, whereas Building remained consistently weak due to visual ambiguity and the mismatch between CLIP's ground-level pre-training and overhead rooftop appearance. The GIS buffer analysis further showed that buffer radius affected both the measured land composition and the classification accuracy, and the cross-dataset evaluation demonstrated that the pipeline generalised best to stable vegetation-dominated scenes while remaining sensitive to construction phases and strong shadow conditions. Overall, the thesis shows that foundation models can support urban tree-centred land composition analysis in a zero-shot setting. Vegetation was estimated reliably across all datasets, road performance improved consistently with distance from the tree centre, and building classification remained unreliable within the small buffer zones, with overall performance strongly influenced by image acquisition conditions and scene structure.

Future work could extend the pipeline in several directions. Evaluating across more cities and sensor types would show how well the pipeline generalises beyond Erlangen. Matched seasonal imagery would also help separate the effects of time, season, and sensor type. Incorporating auxiliary spatial data such as cadastral building footprints or digital surface models could improve building classification and shadow handling, which were the main observed failure modes. On the segmentation side, analysing the direct relationship between segmentation structure and downstream classification accuracy would support more comprehensive pipeline optimisation. Since existing remote-sensing CLIP variants are generally finetuned on image-level data, finetuning on segment-level labelled data would likely yield more discriminative embeddings for urban land cover classes, particularly for visually ambiguous categories such as buildings and roads. Finally, extending the GIS buffer analysis to a larger tree inventory and comparing the estimated land composition against field-based urban greening indicators would help assess the practical value of the approach for urban planning and environmental monitoring.

List of Figures

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